

Rqc 1.14.0 - Quality Control Report

File Information

This table describes input files. `reads` column can be total number of reads (`sample=FALSE`) or sample size.

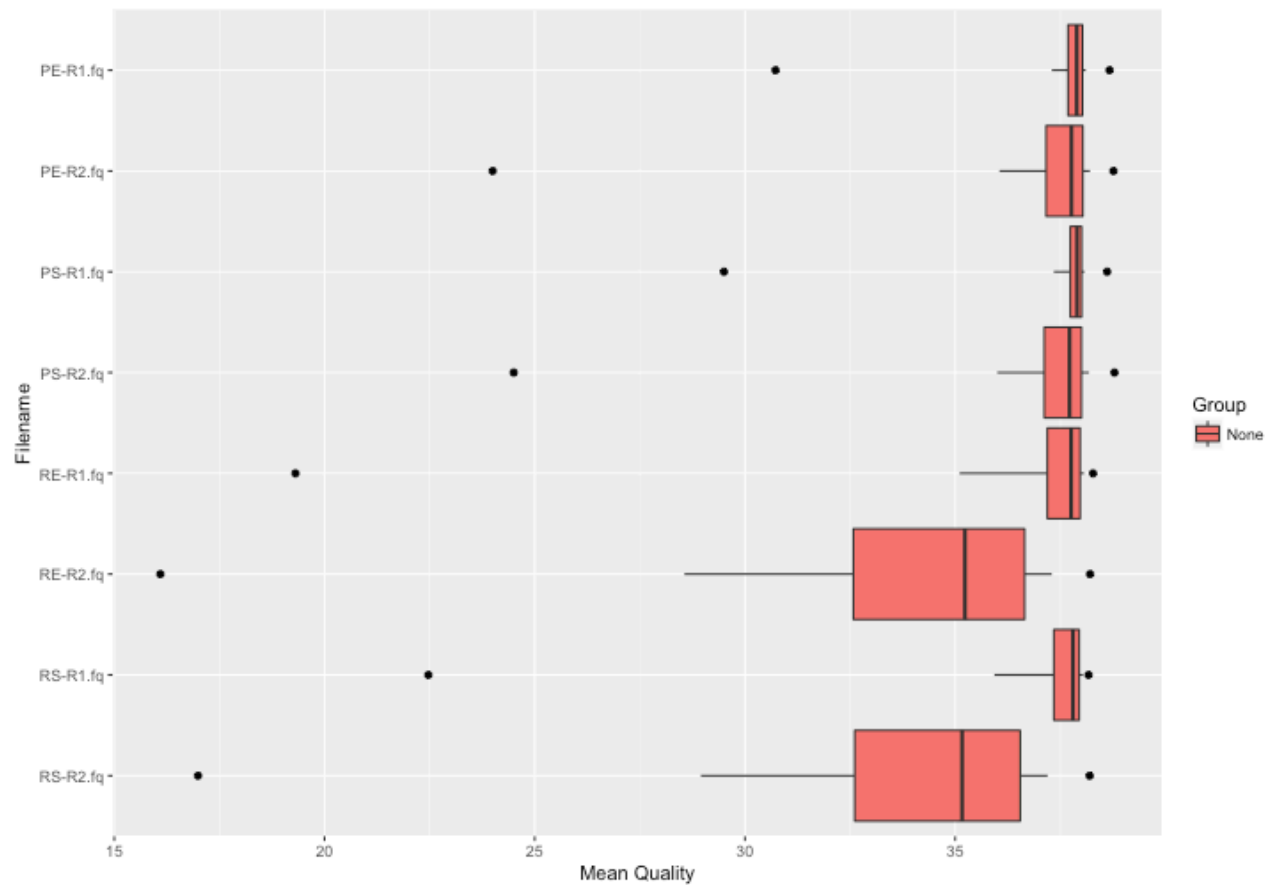
```
kable(perFileInformation(rqcResultSet))
```

filename	pair	format	group	reads	total.reads	path
PE-R1.fq	1	FASTQ	None	38536	38536	/Users/morxo28p/Micro336-master/data
PE-R2.fq	2	FASTQ	None	38536	38536	/Users/morxo28p/Micro336-master/data
PS-R1.fq	3	FASTQ	None	19113	19113	/Users/morxo28p/Micro336-master/data
PS-R2.fq	4	FASTQ	None	19113	19113	/Users/morxo28p/Micro336-master/data
RE-R1.fq	5	FASTQ	None	38537	38537	/Users/morxo28p/Micro336-master/data
RE-R2.fq	6	FASTQ	None	38537	38537	/Users/morxo28p/Micro336-master/data
RS-R1.fq	7	FASTQ	None	19113	19113	/Users/morxo28p/Micro336-master/data
RS-R2.fq	8	FASTQ	None	19113	19113	/Users/morxo28p/Micro336-master/data

Per Read Mean Quality Distribution of Files

This plot describe an overview of per read mean quality distribution of all files

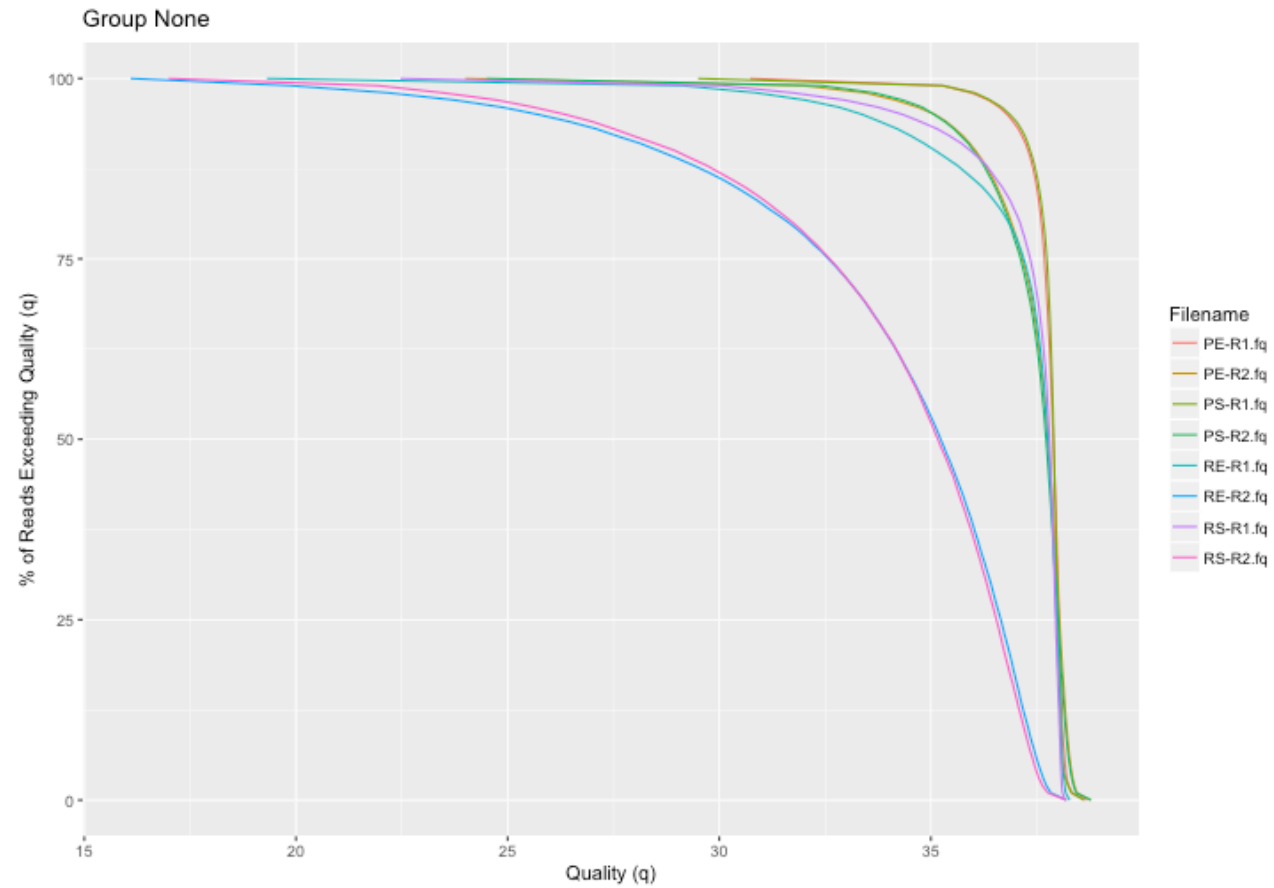
```
rqcReadQualityBoxPlot(rqcResultSet)
```



Average Quality

This plot describes the average quality pattern by showing on the X-axis quality thresholds and on the Y-axis the percentage of reads that exceed that quality level.

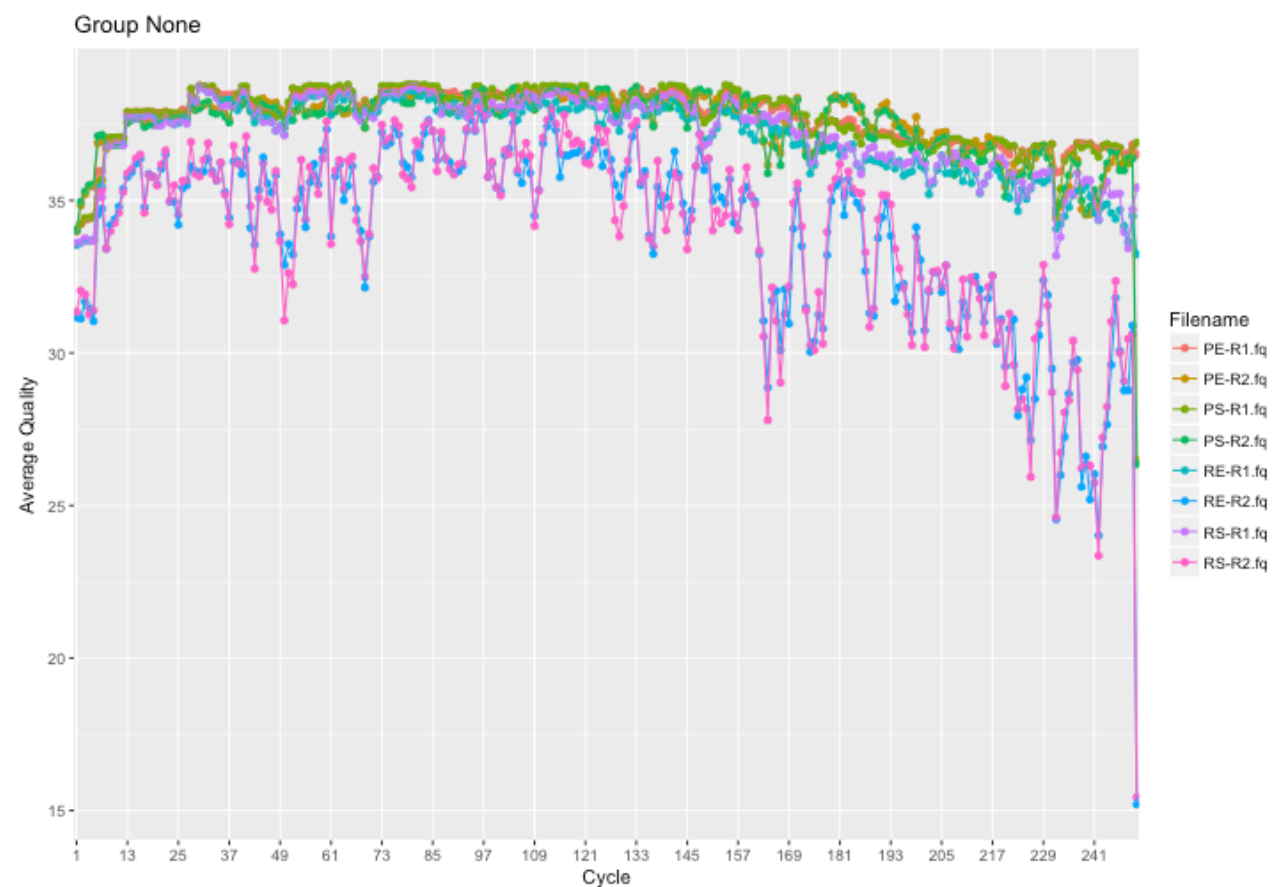
```
groups <- unique(perFileInformation(rqcResultSet)$group)
for (group in groups) {
  rqcResultSet.sub <- subsetByGroup(rqcResultSet, group)
  print(rqcReadQualityPlot(rqcResultSet.sub) +
        ggtitle(paste("Group", group)))
}
```



Cycle-specific Average Quality

This plot describes the average quality scores for each cycle of sequencing.

```
for (group in groups) {
  rqcResultSet.sub <- subsetByGroup(rqcResultSet, group)
  print(rqcCycleAverageQualityPlot(rqcResultSet.sub) +
        ggtitle(paste("Group", group)))
}
```



PCA Biplot (cycle-specific read average quality)

Biplot from Principal Component Analysis (PCA) of cycle-specific read average quality.

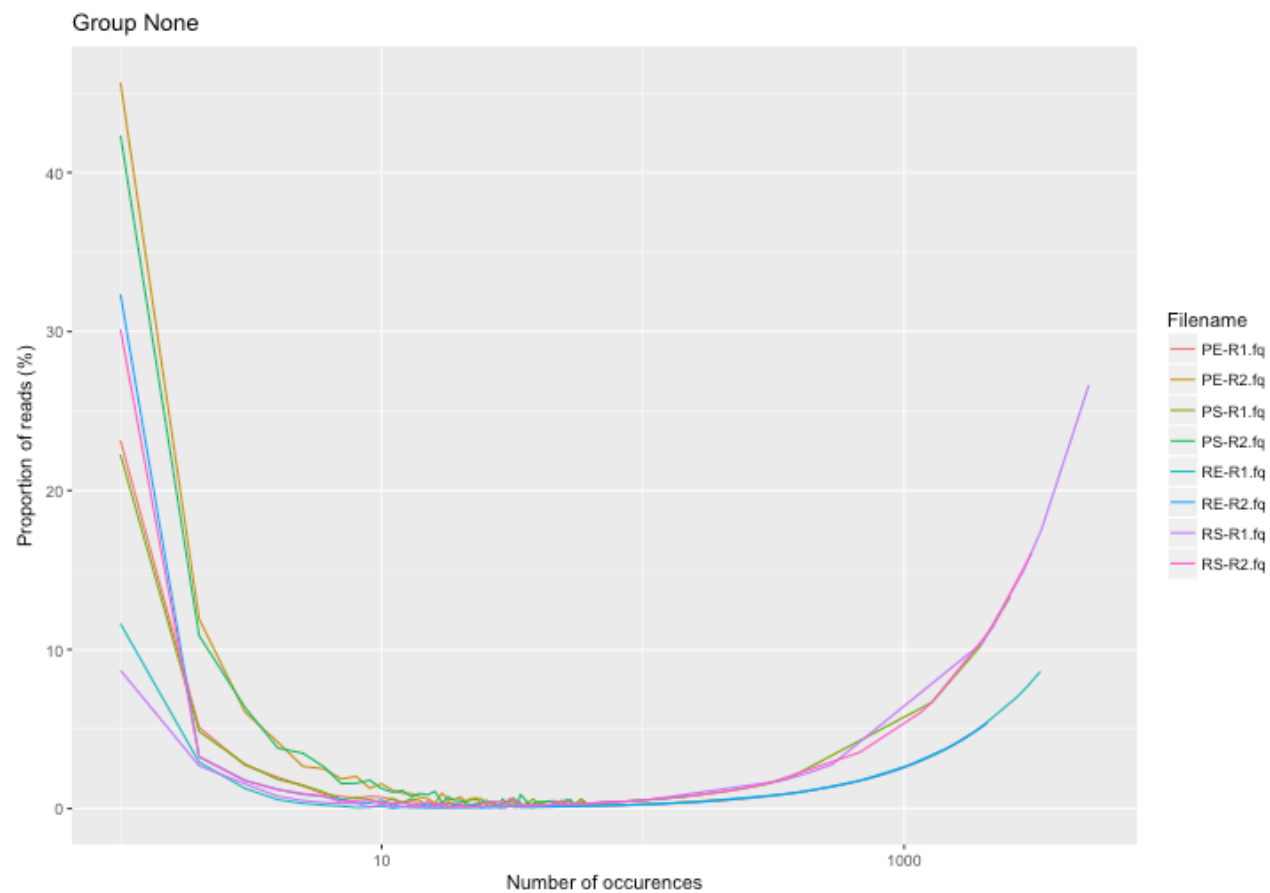
```
for (group in groups) {  
  rqcResultSet.sub <- subsetByGroup(rqcResultSet, group)  
  print(rqcCycleAverageQualityPcaPlot(rqcResultSet.sub) +  
        ggtitle(paste("Group", group)))  
}
```

```
## Error in `[.data.frame`(mf, , x): undefined columns selected
```

Read Frequency

This plot shows the proportion of reads that appeared many times.

```
for (group in groups) {  
  rqcResultSet.sub <- subsetByGroup(rqcResultSet, group)  
  print(rqcReadFrequencyPlot(rqcResultSet.sub) +  
        ggtitle(paste("Group", group)))  
}
```



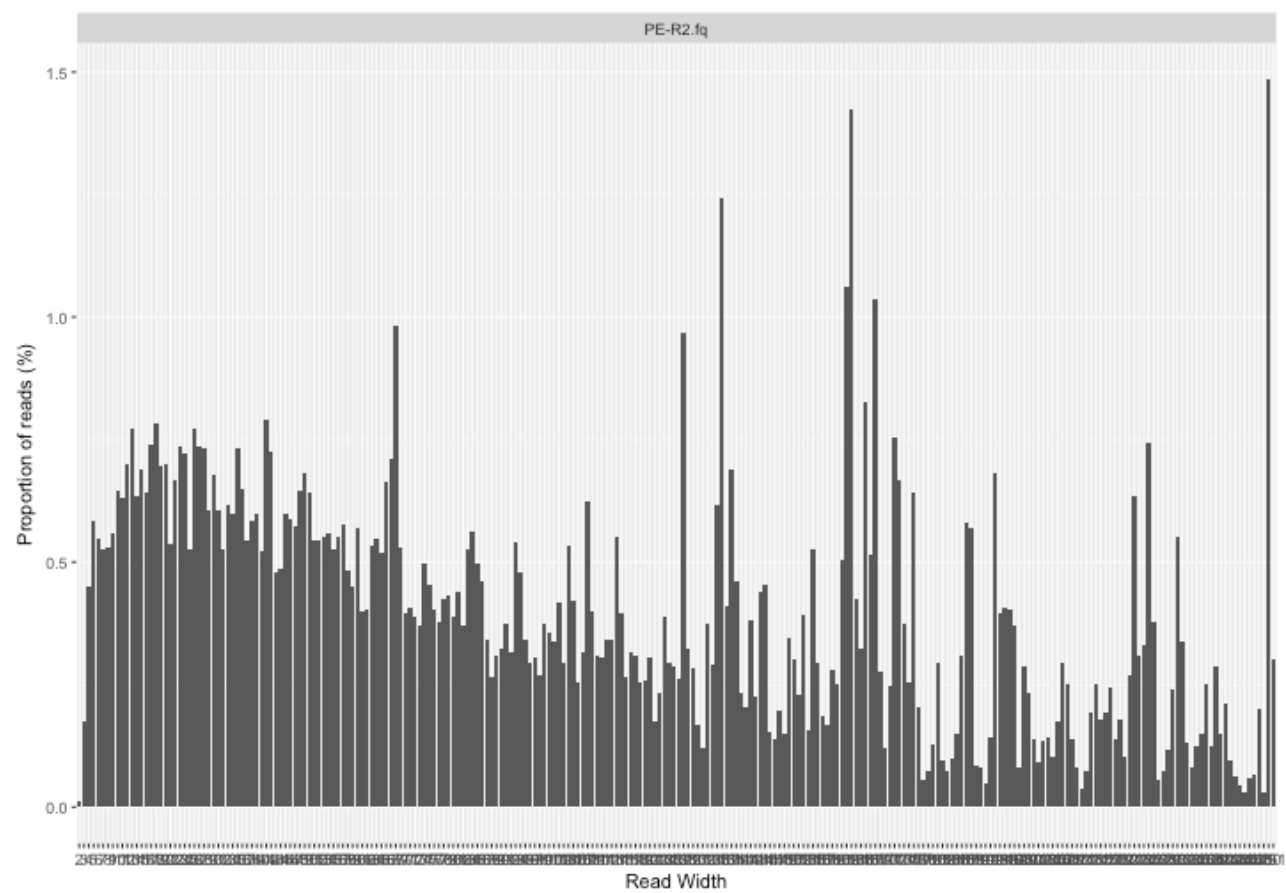
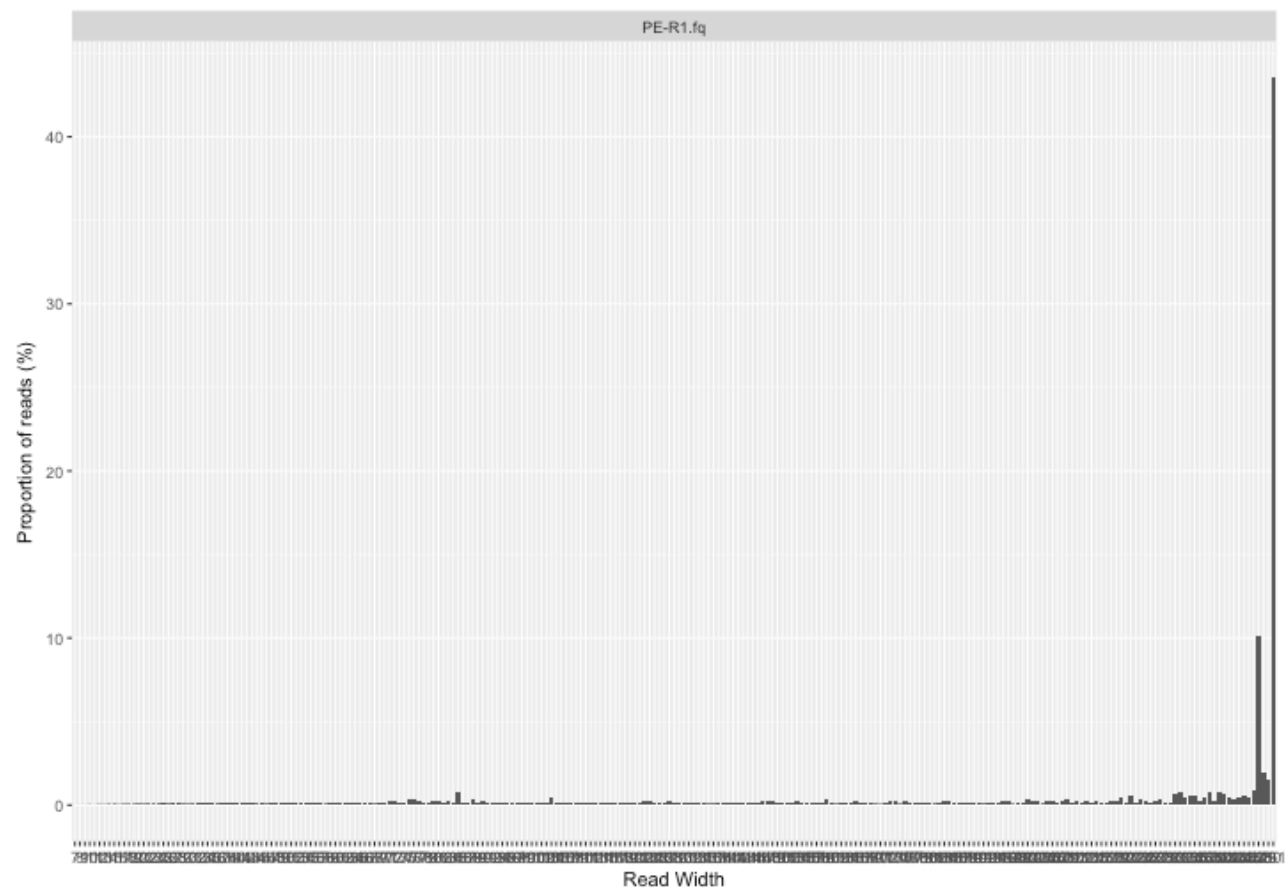
Read Length Distribution

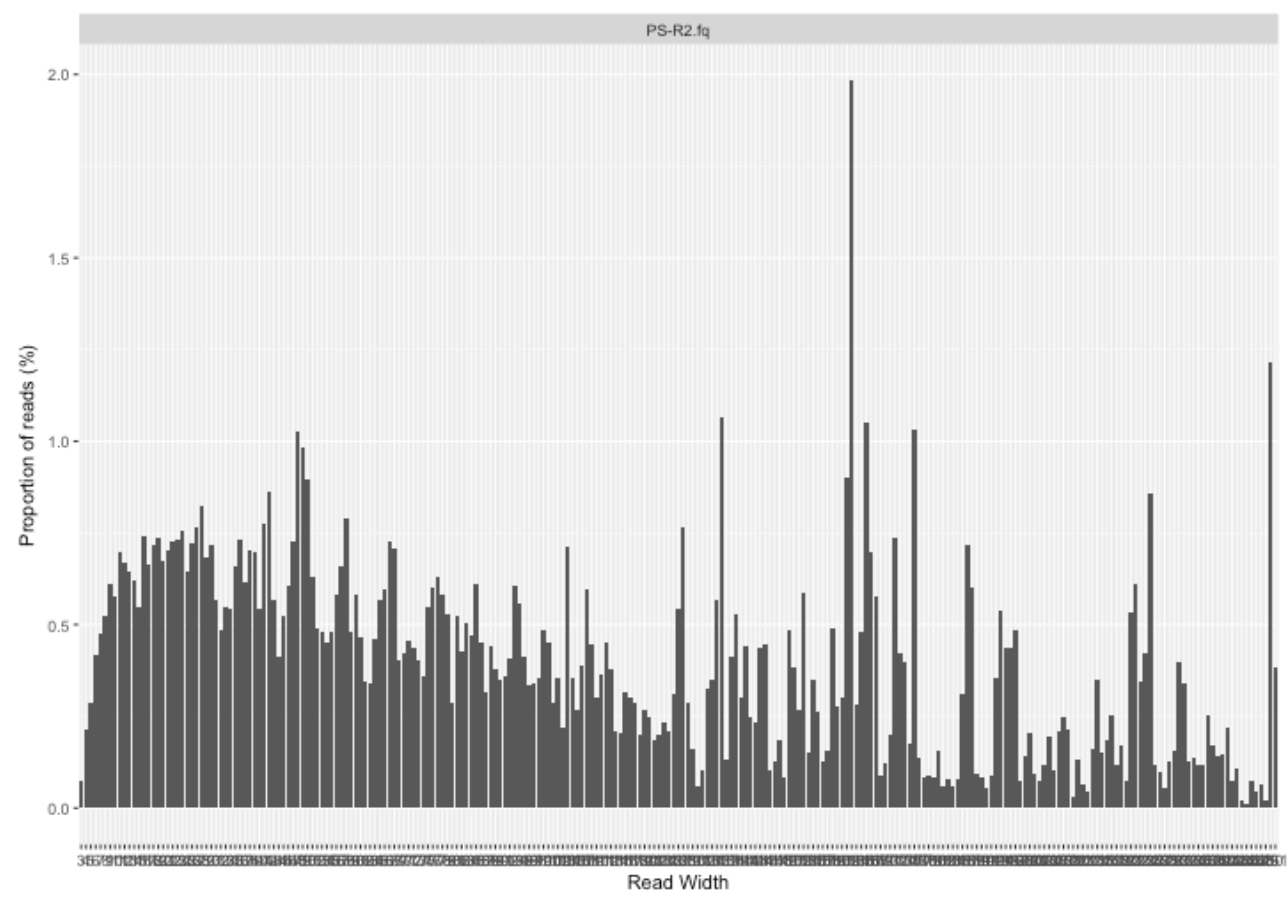
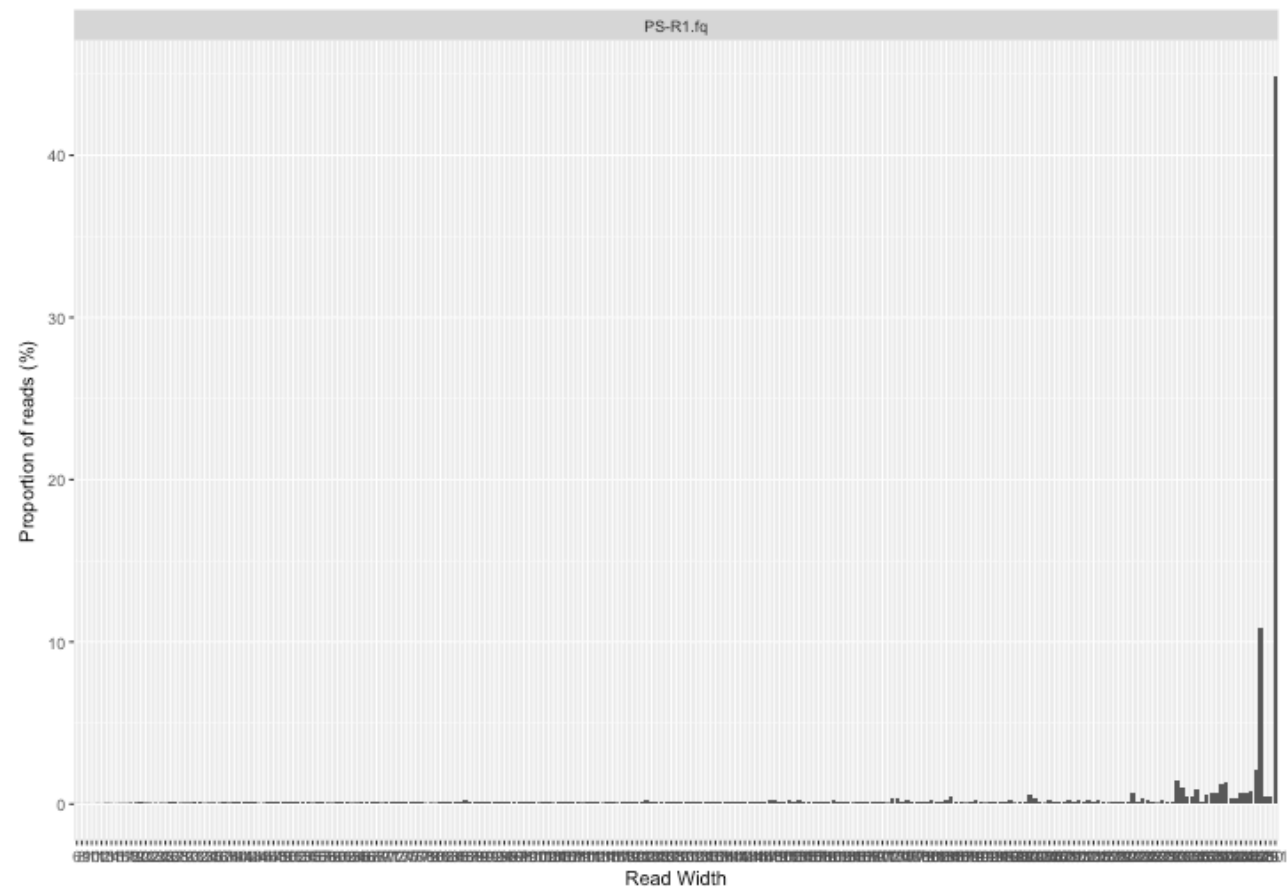
Barplot that presents the distribution of the lengths of the reads available in the FASTQ file.

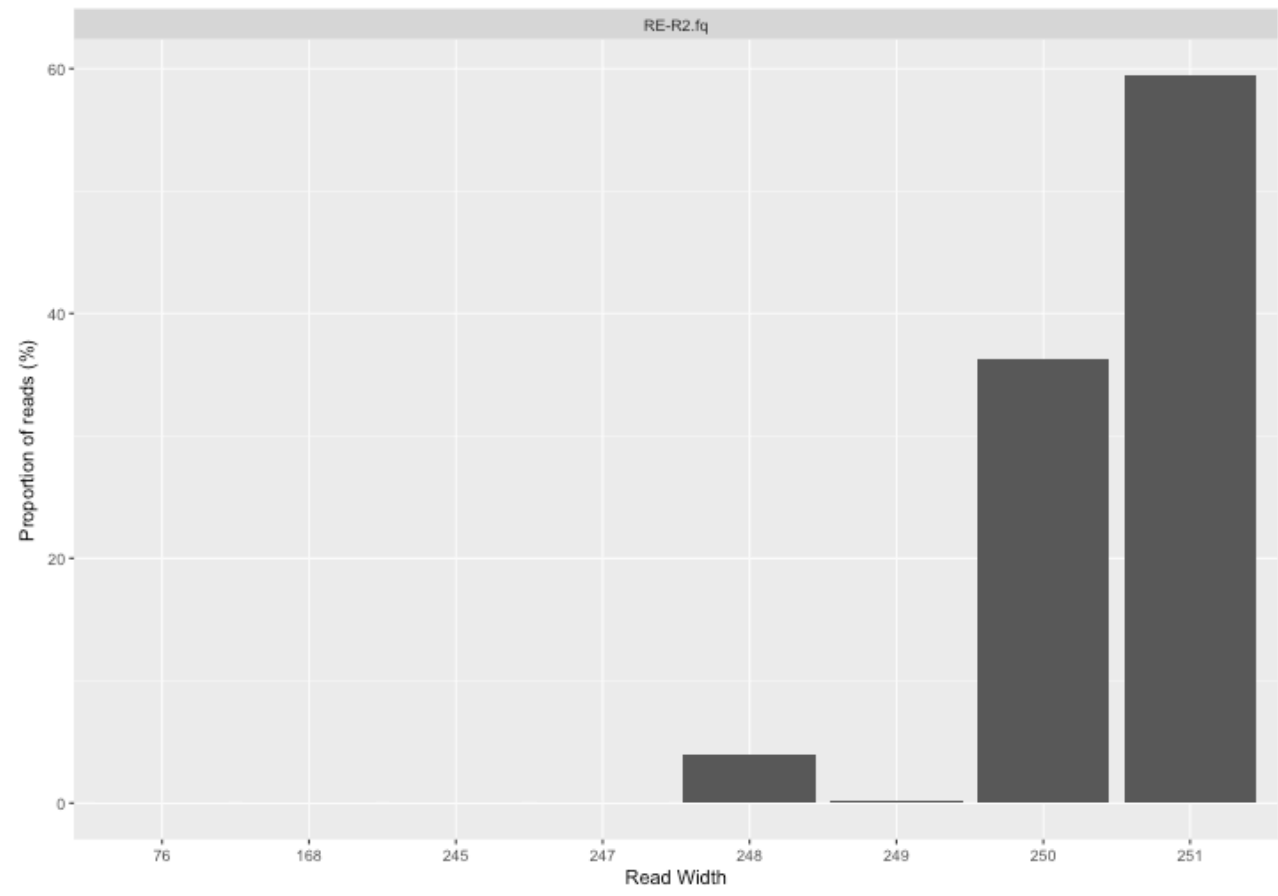
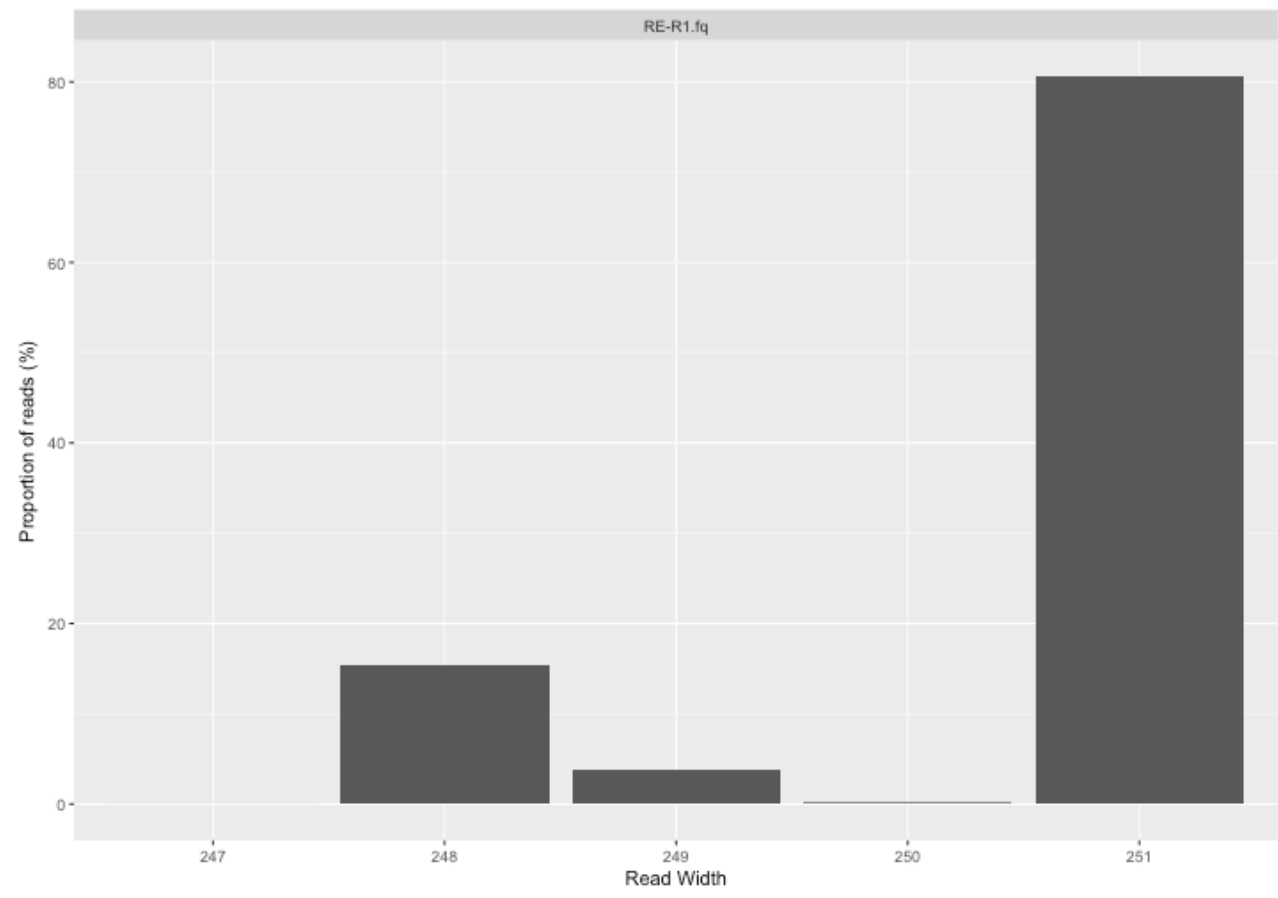
```

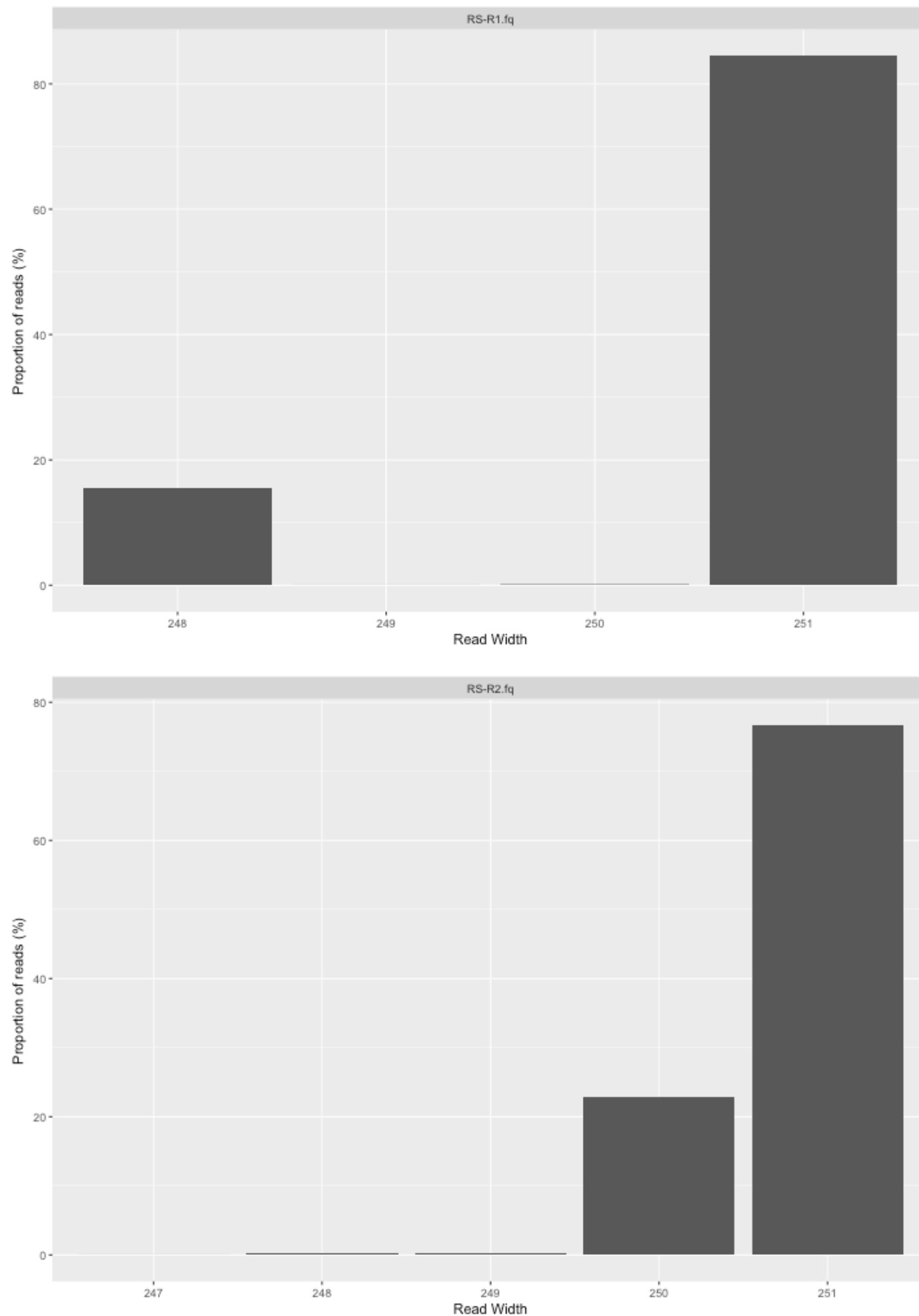
pairs <- unique(perFileInformation(rqcResultSet)$pair)
for(pair in pairs) {
  rqcResultSet.sub <- subsetByPair(rqcResultSet, pair)
  print(rqcReadWidthPlot(rqcResultSet.sub))
}

```





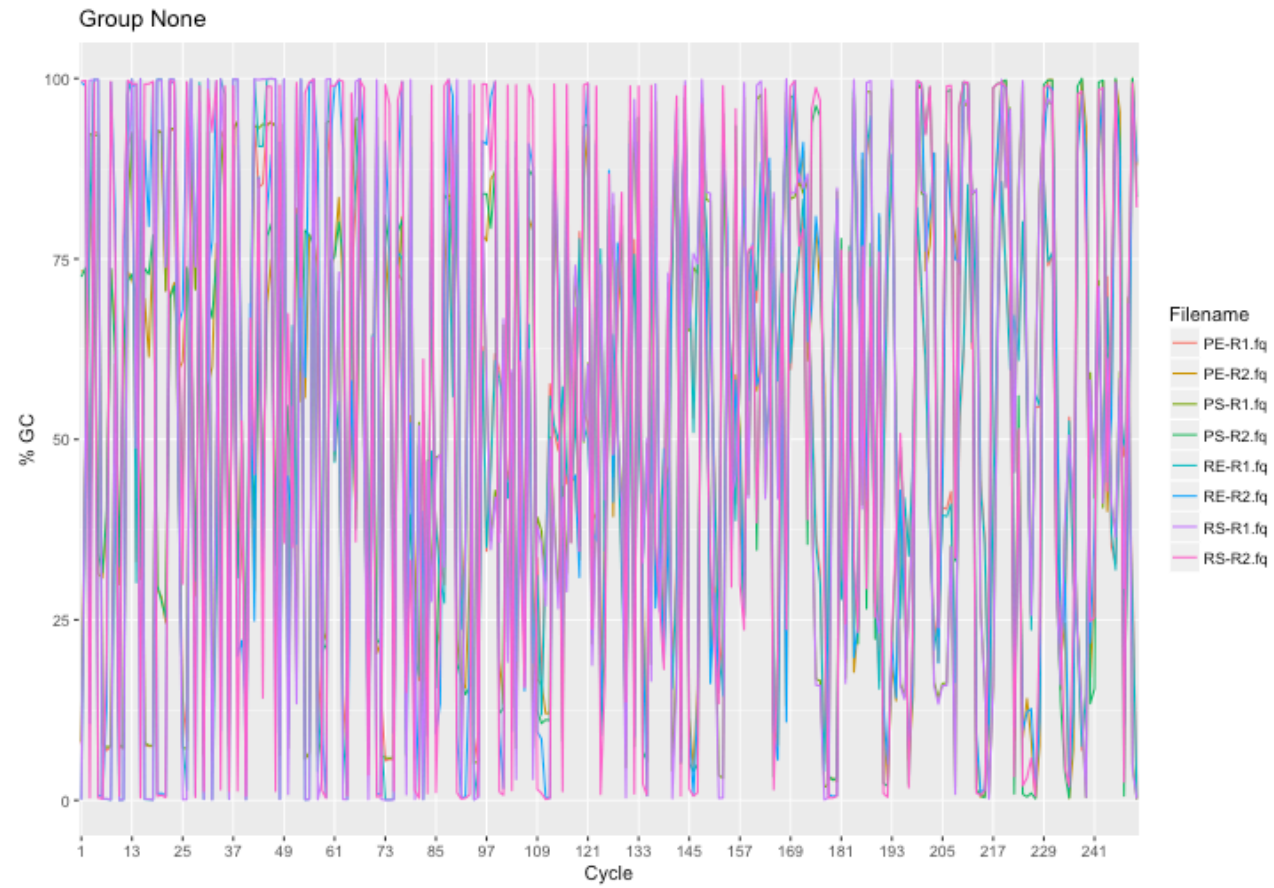




Cycle-specific GC Content

Line plot showing the average GC content for every cycle of sequencing.

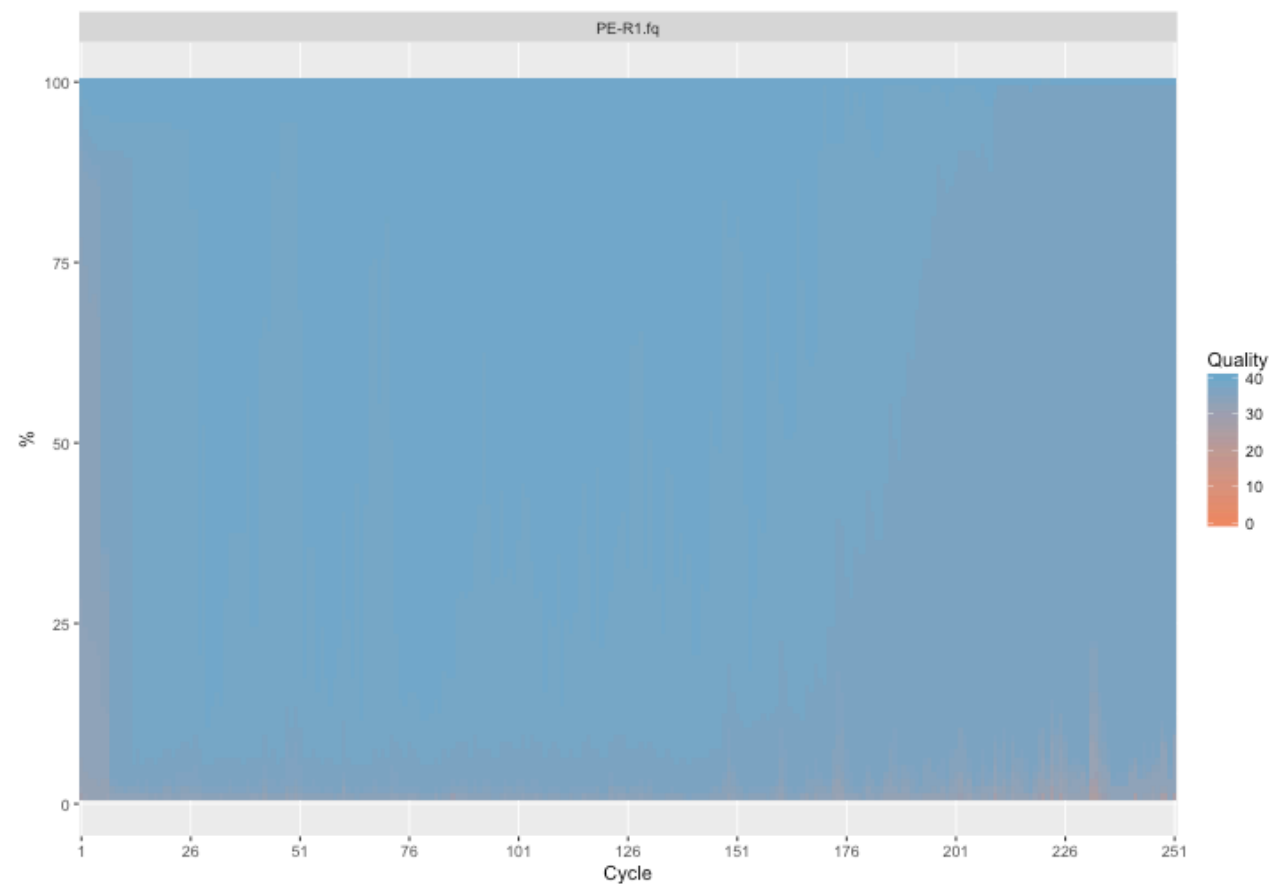
```
for (group in groups) {  
  rqcResultSet.sub <- subsetByGroup(rqcResultSet, group)  
  print(rqcCycleGCPlot(rqcResultSet.sub) +  
        ggtitle(paste("Group", group)))  
}
```

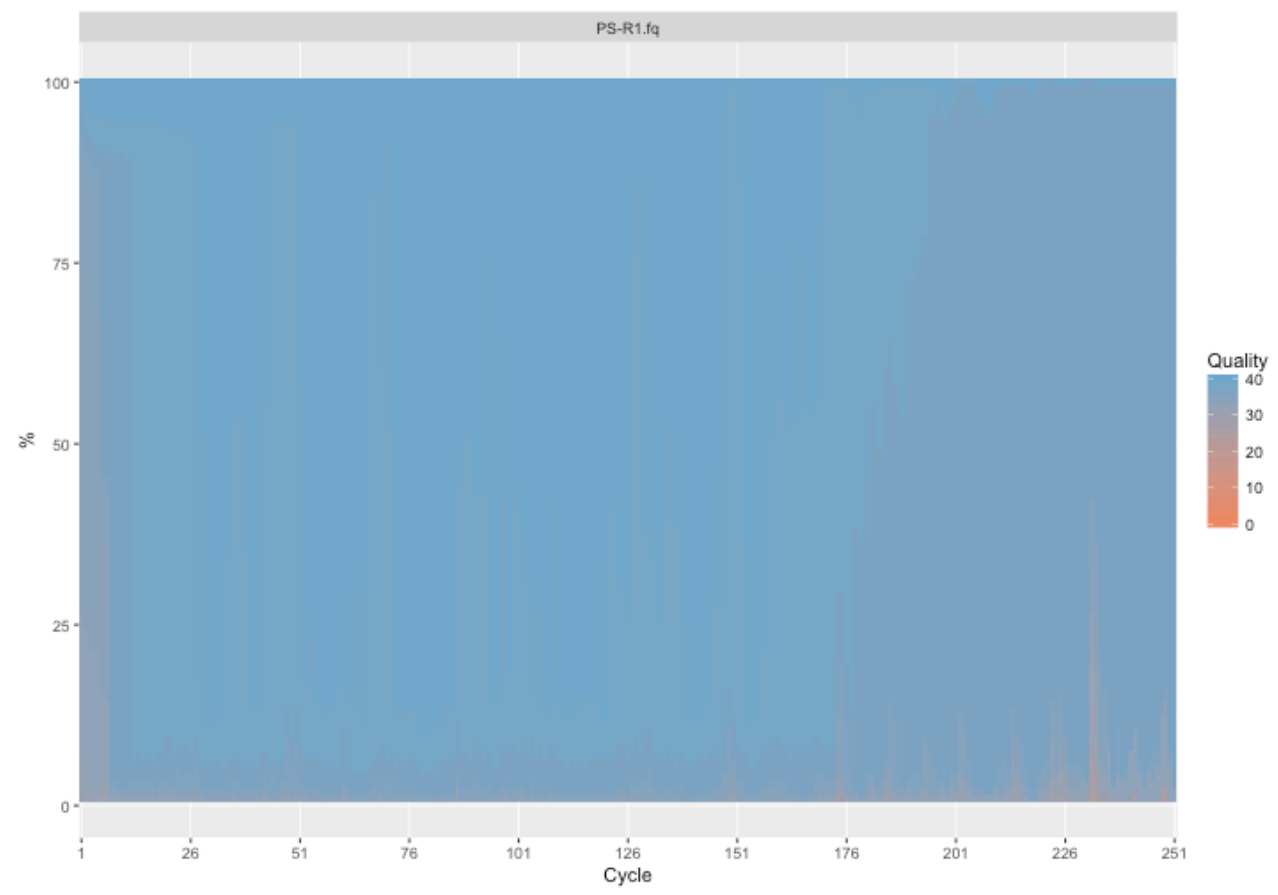
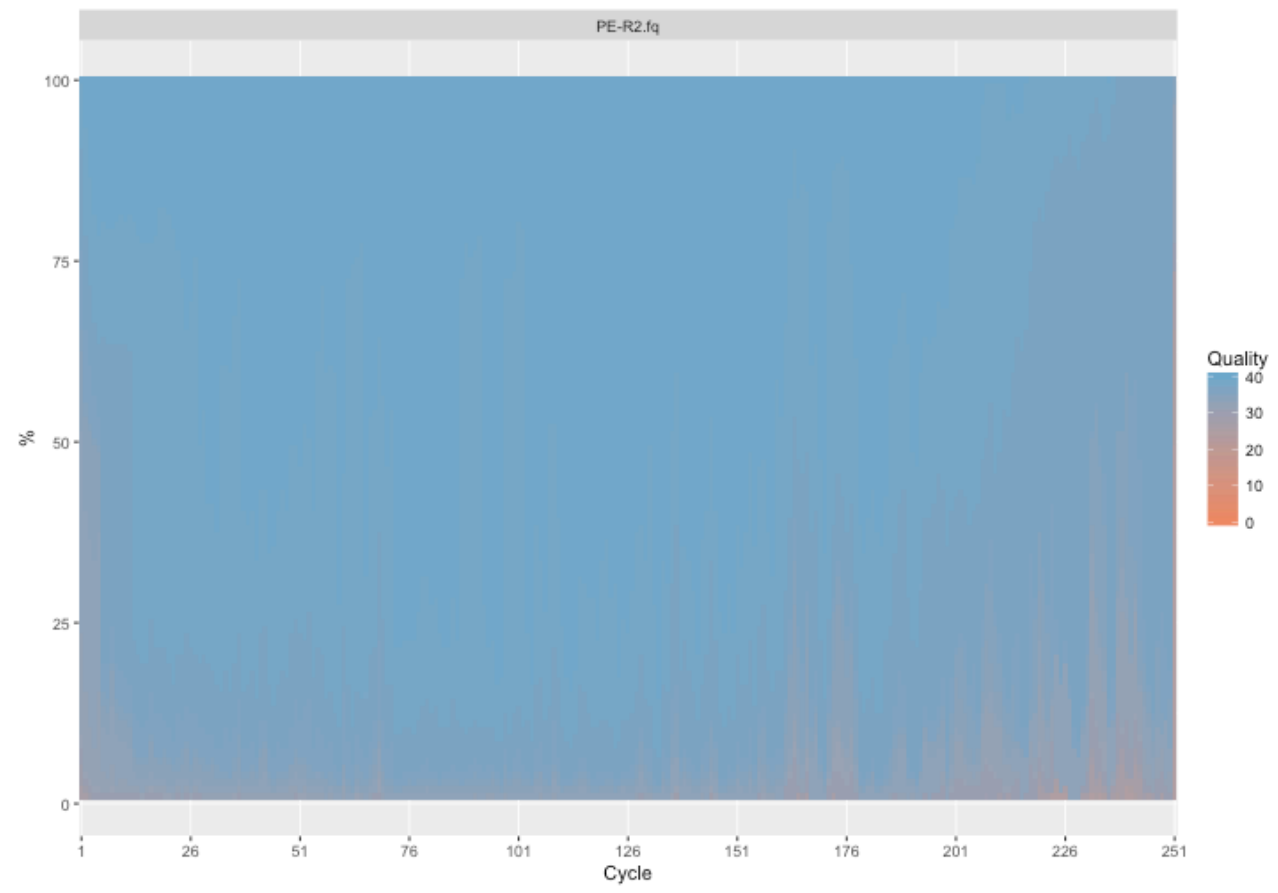


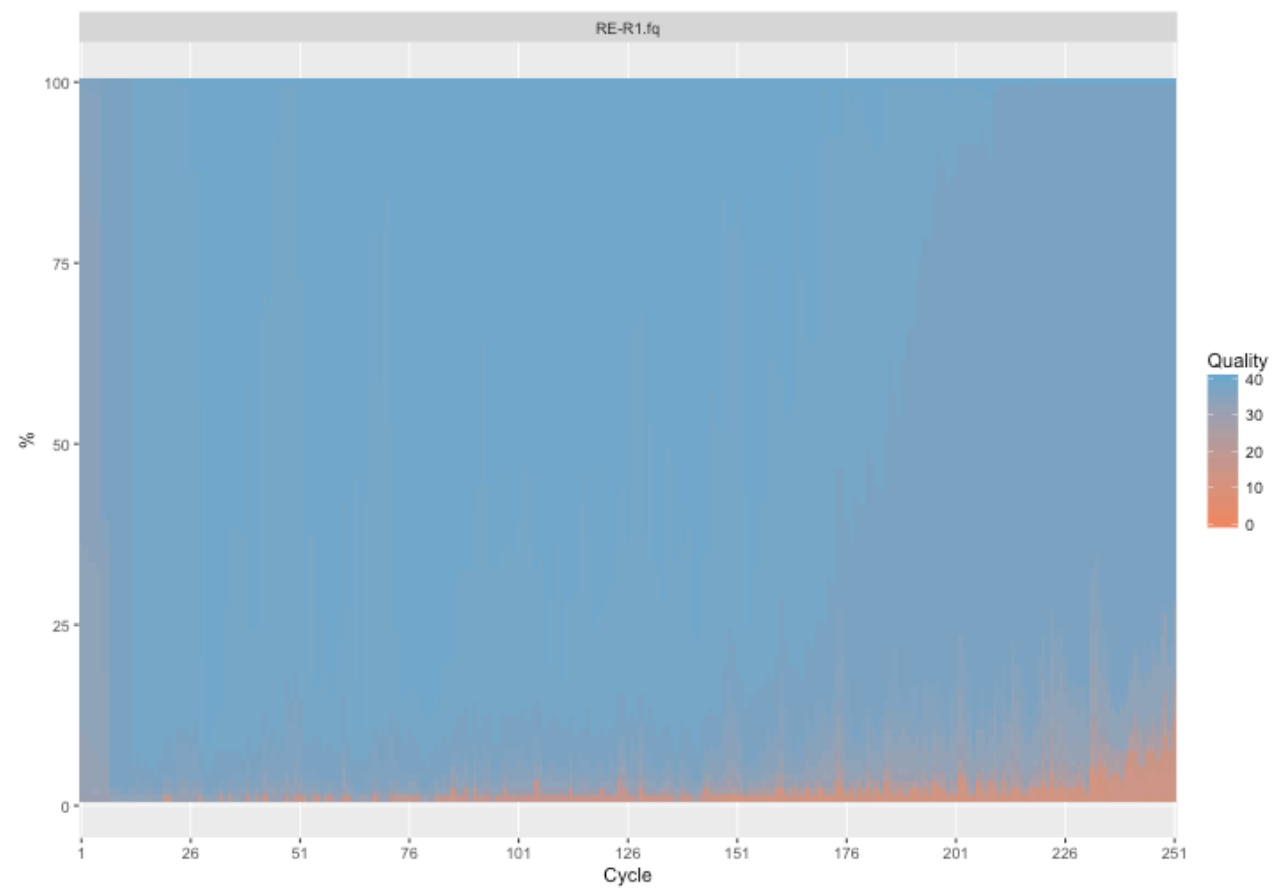
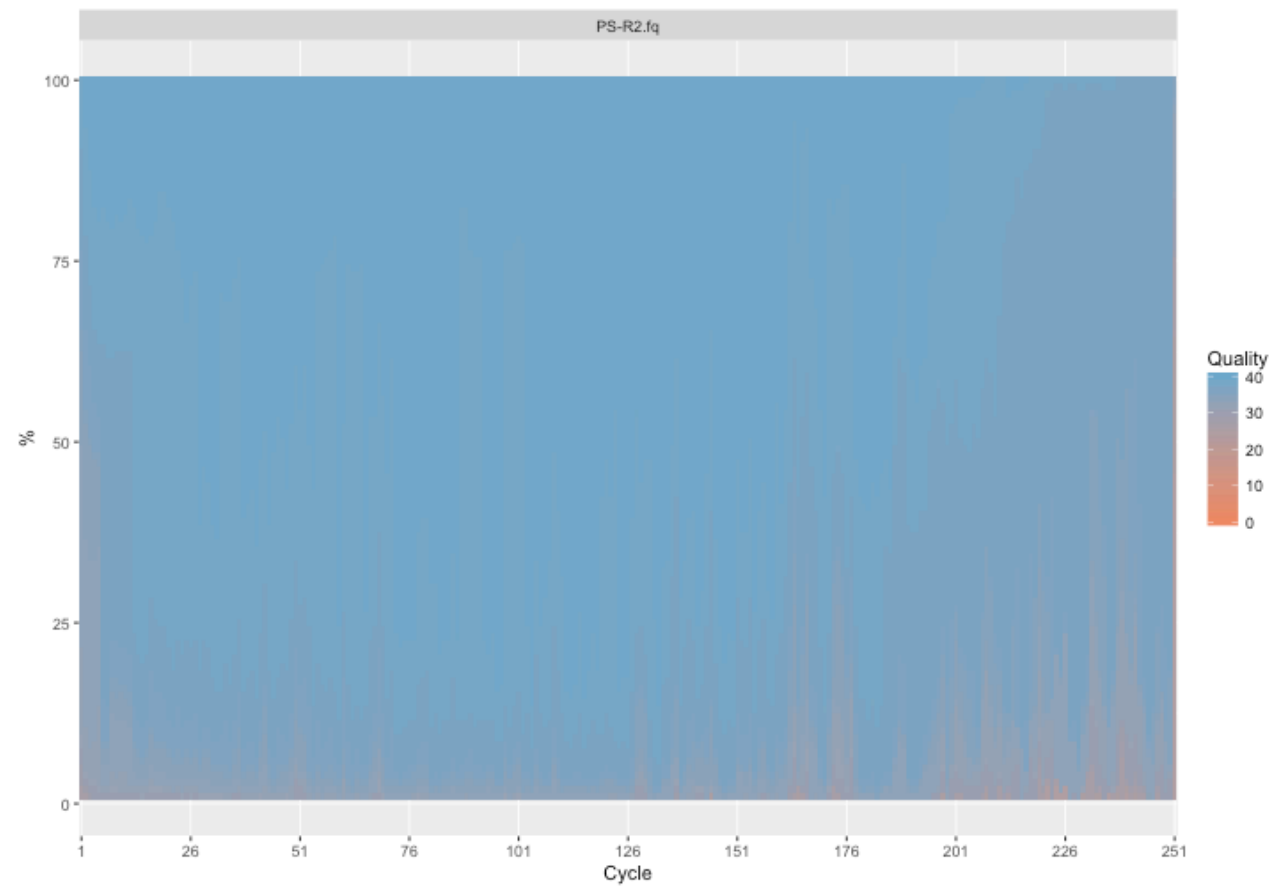
Cycle-specific Quality Distribution

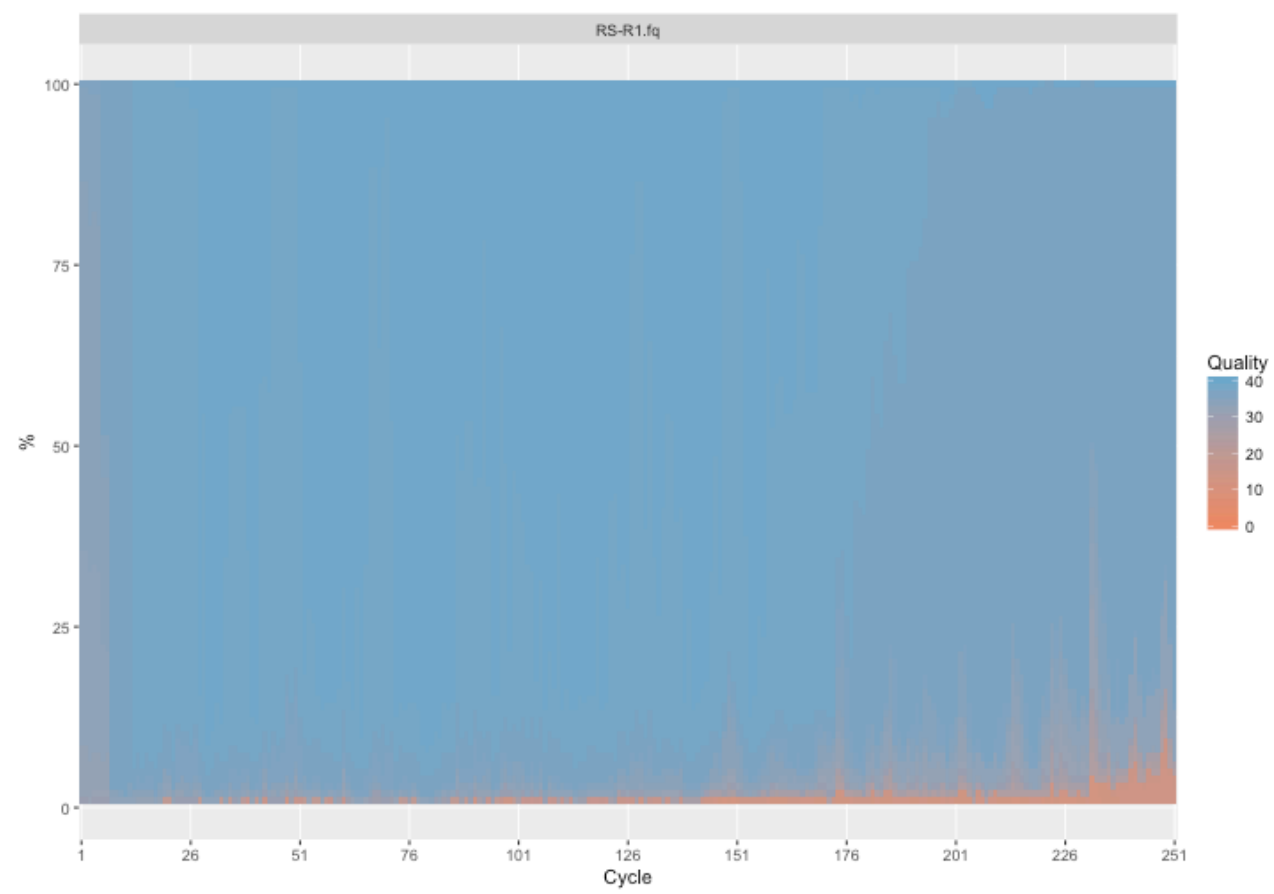
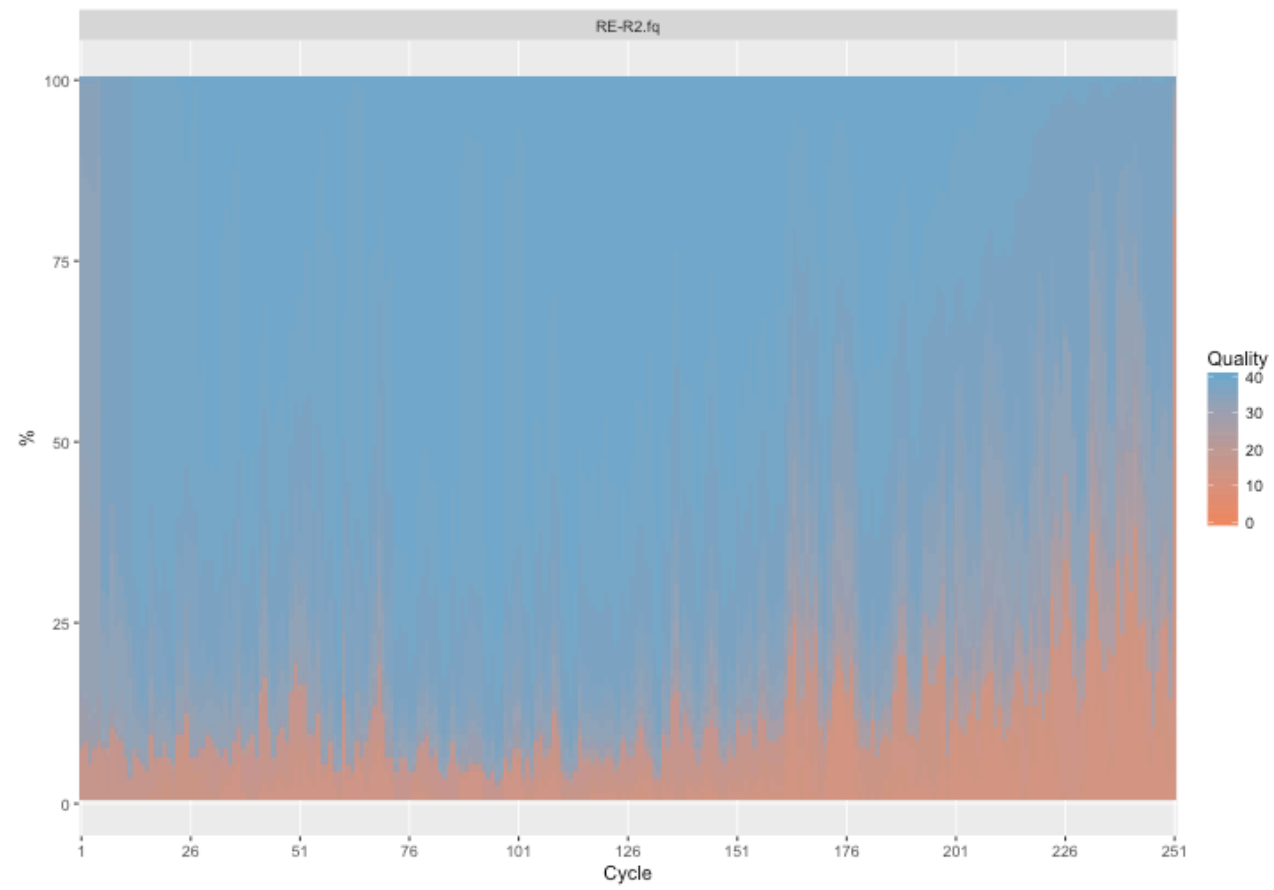
Bar plot showing the proportion of quality calls per cycle. Colors are presented in a gradient Red-Blue, where red identifies calls of lower quality. This visualization is preferred as it is cleaner than the boxplots described below.

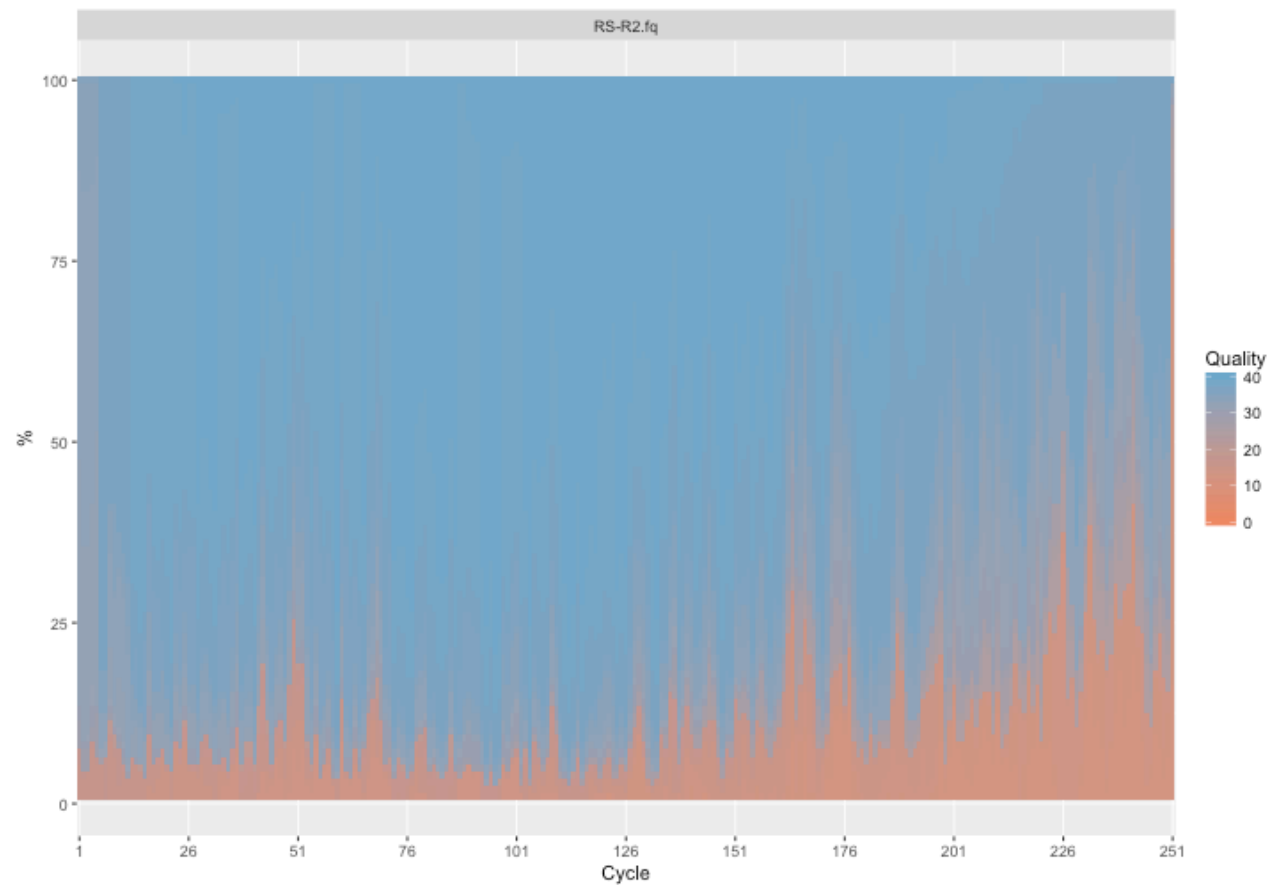
```
for(pair in pairs) {
  rqcResultSet.sub <- subsetByPair(rqcResultSet, pair)
  print(rqcCycleQualityPlot(rqcResultSet.sub))
}
```







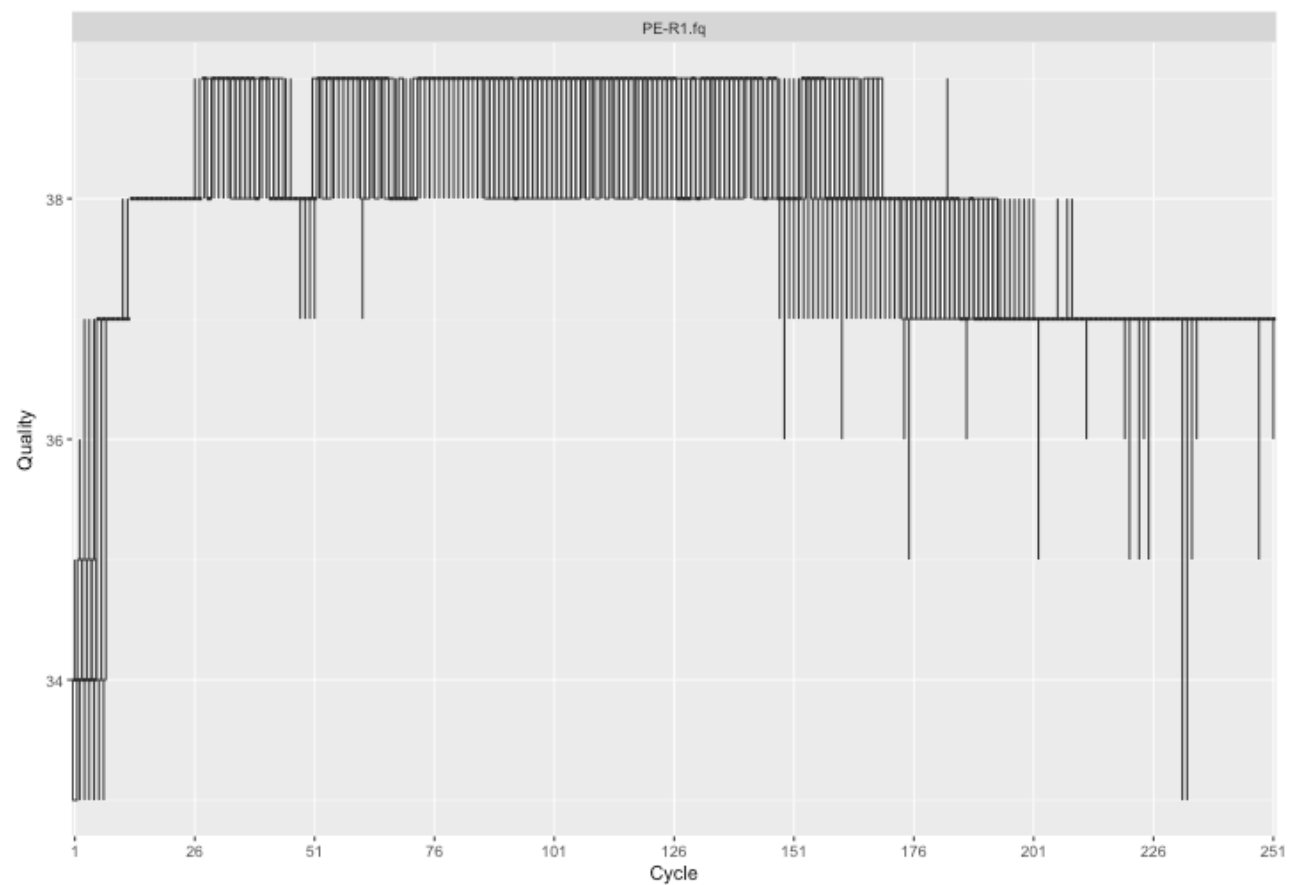


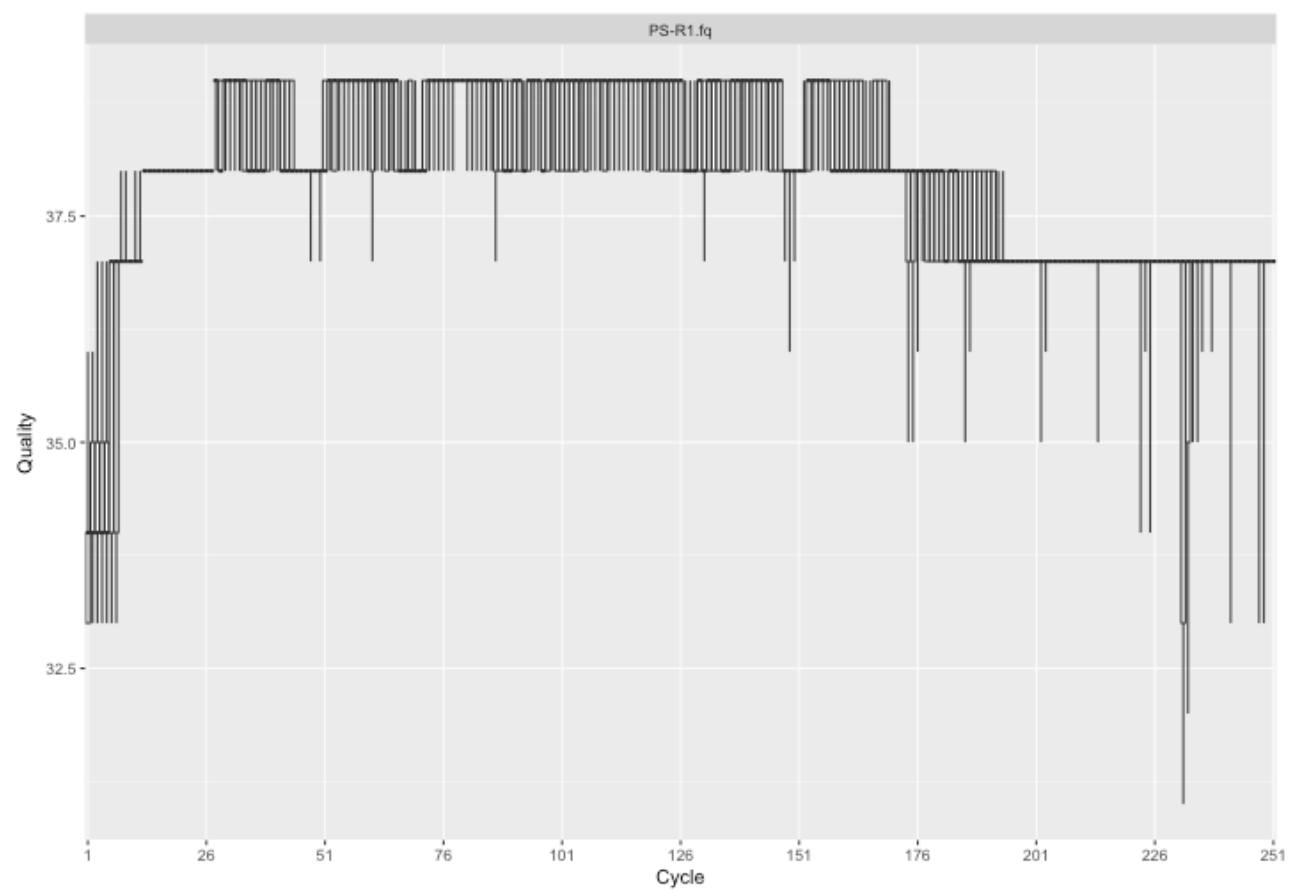
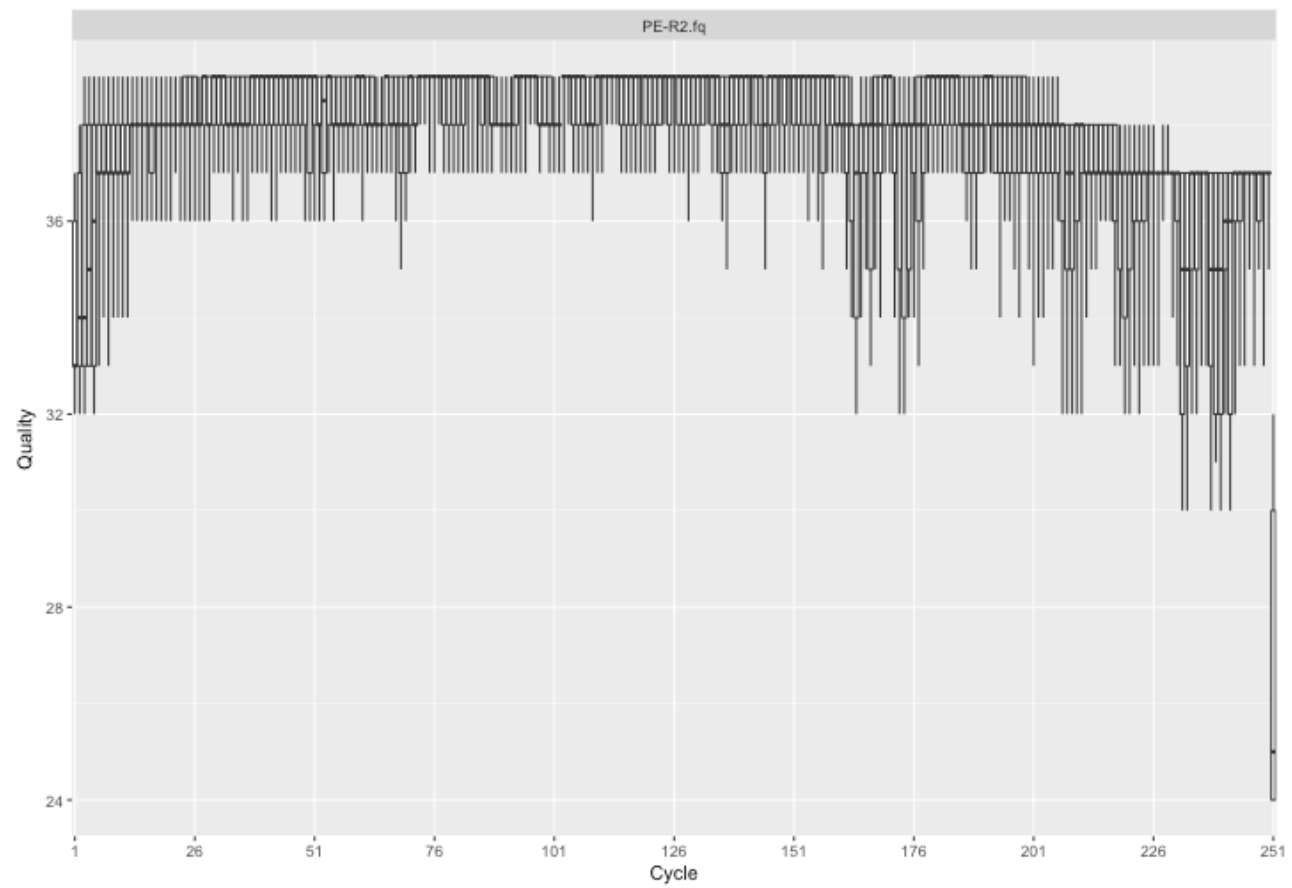


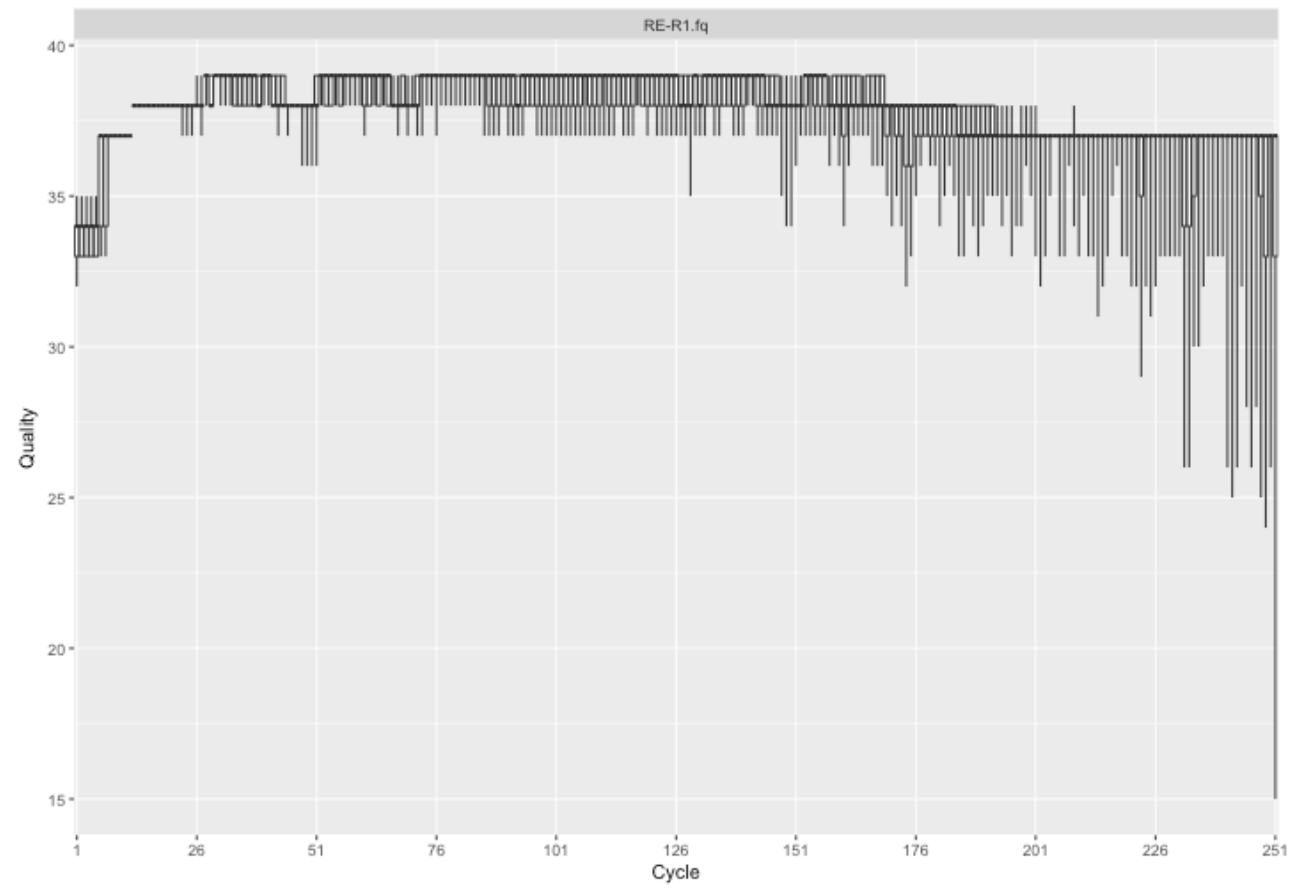
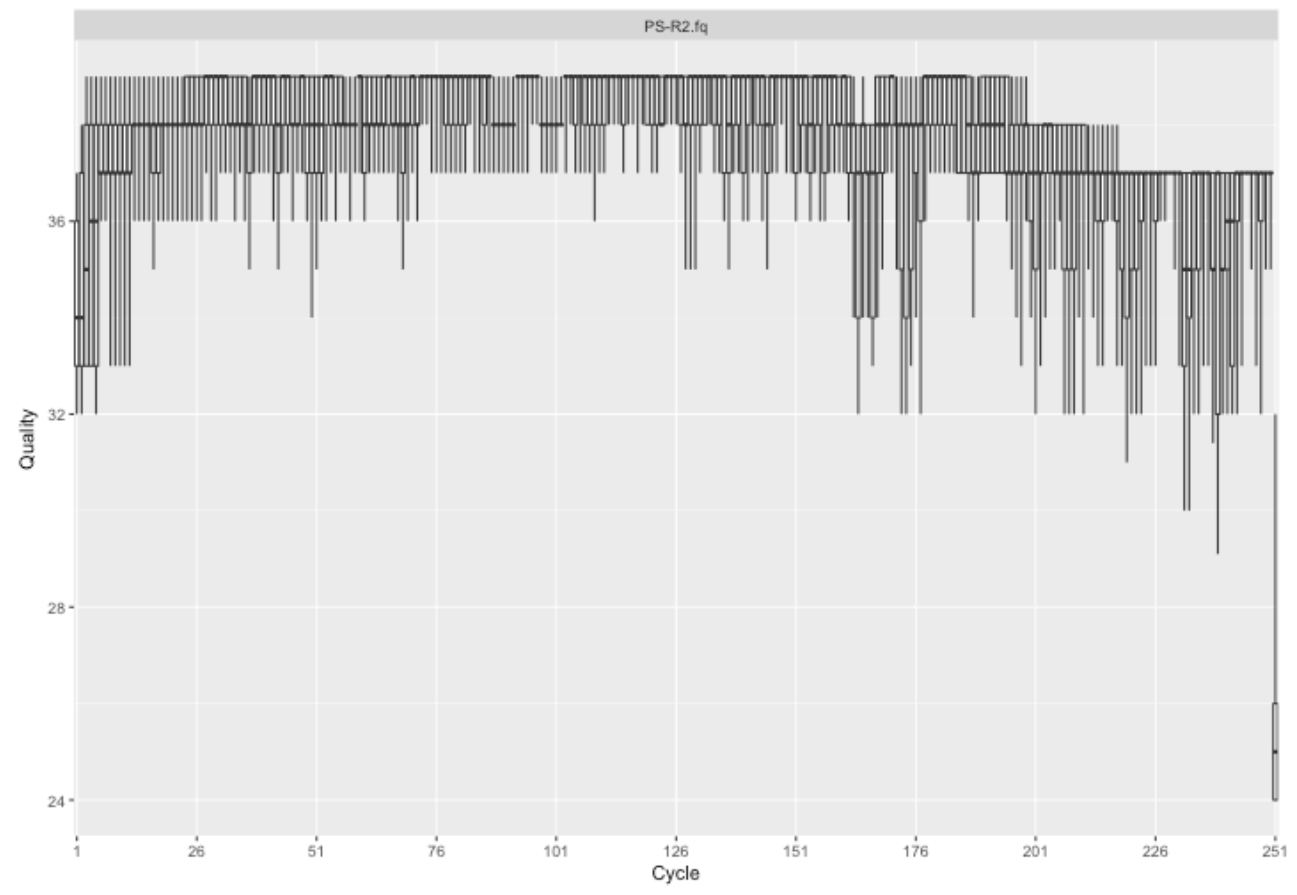
Cycle-specific Quality Distribution - Boxplot

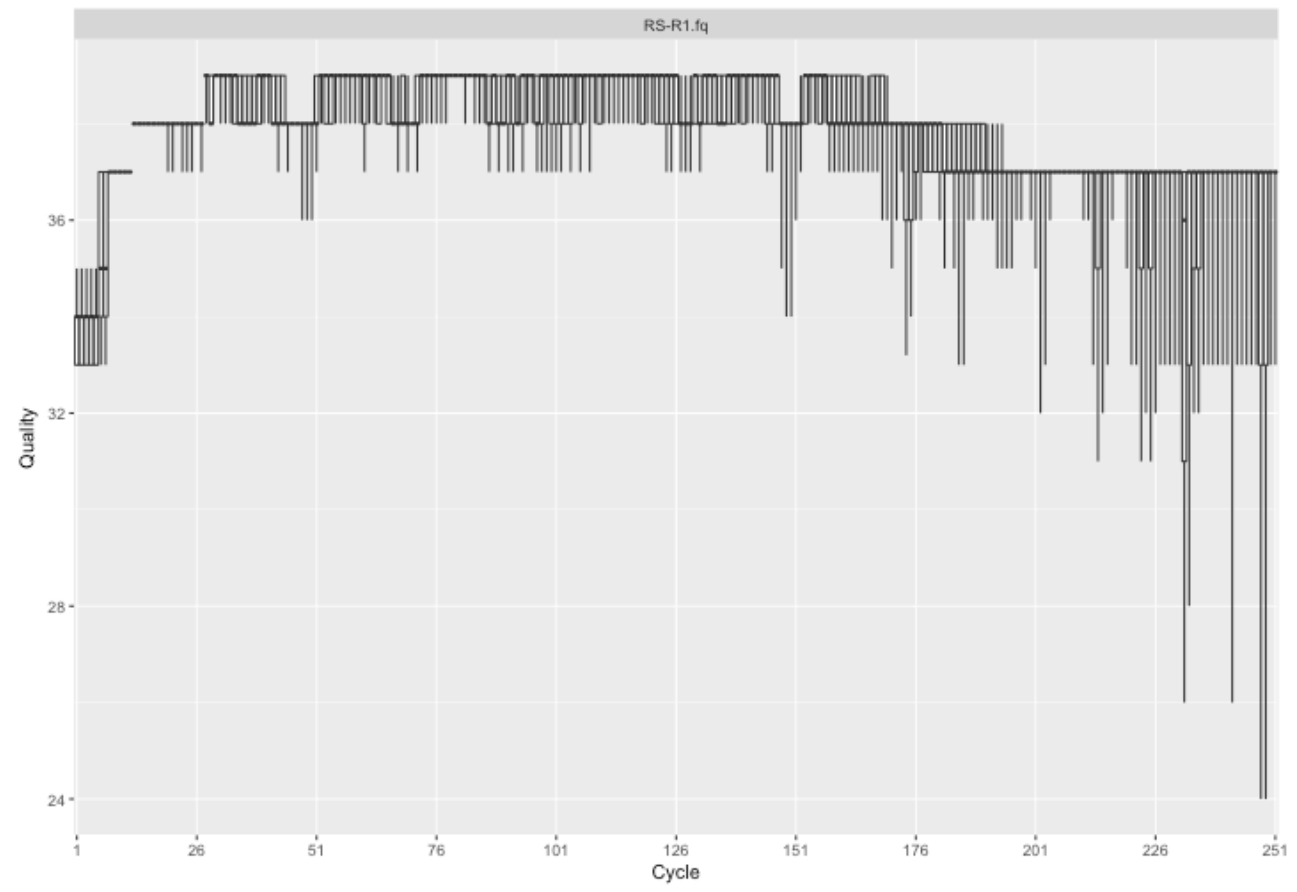
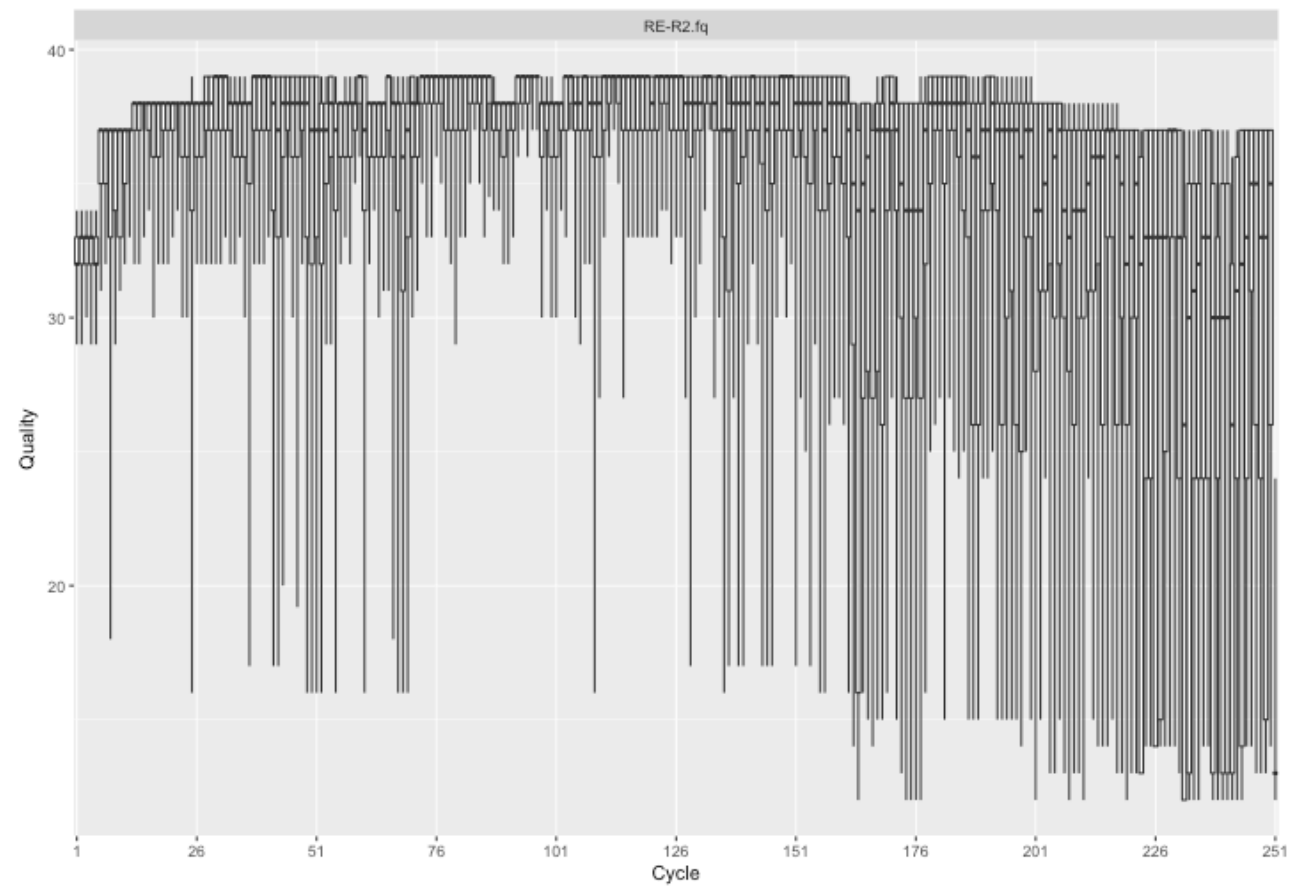
Boxplots describing empirical patterns of quality distribution on each cycle of sequencing.

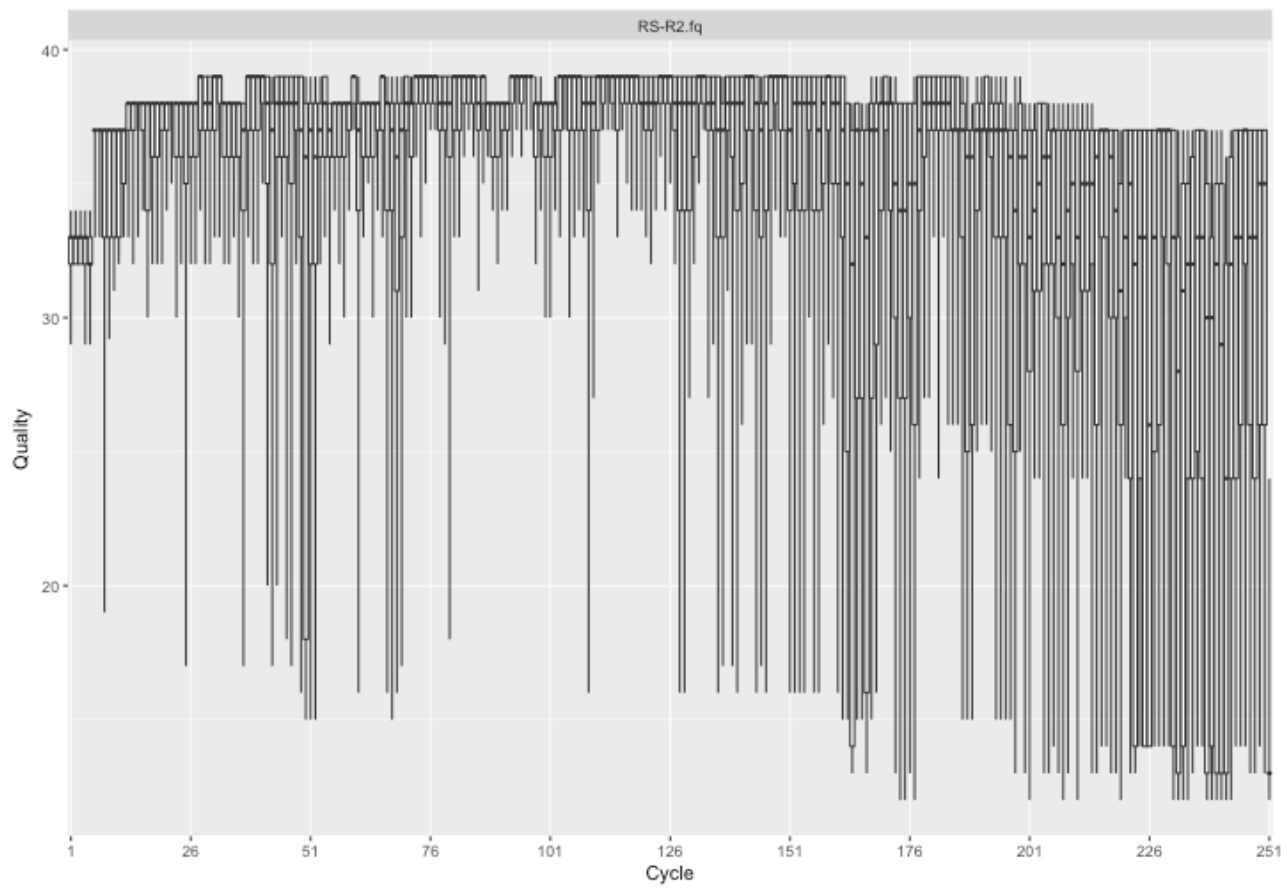
```
for(pair in pairs) {
  rqcResultSet.sub <- subsetByPair(rqcResultSet, pair)
  print(rqcCycleQualityBoxPlot(rqcResultSet.sub))
}
```







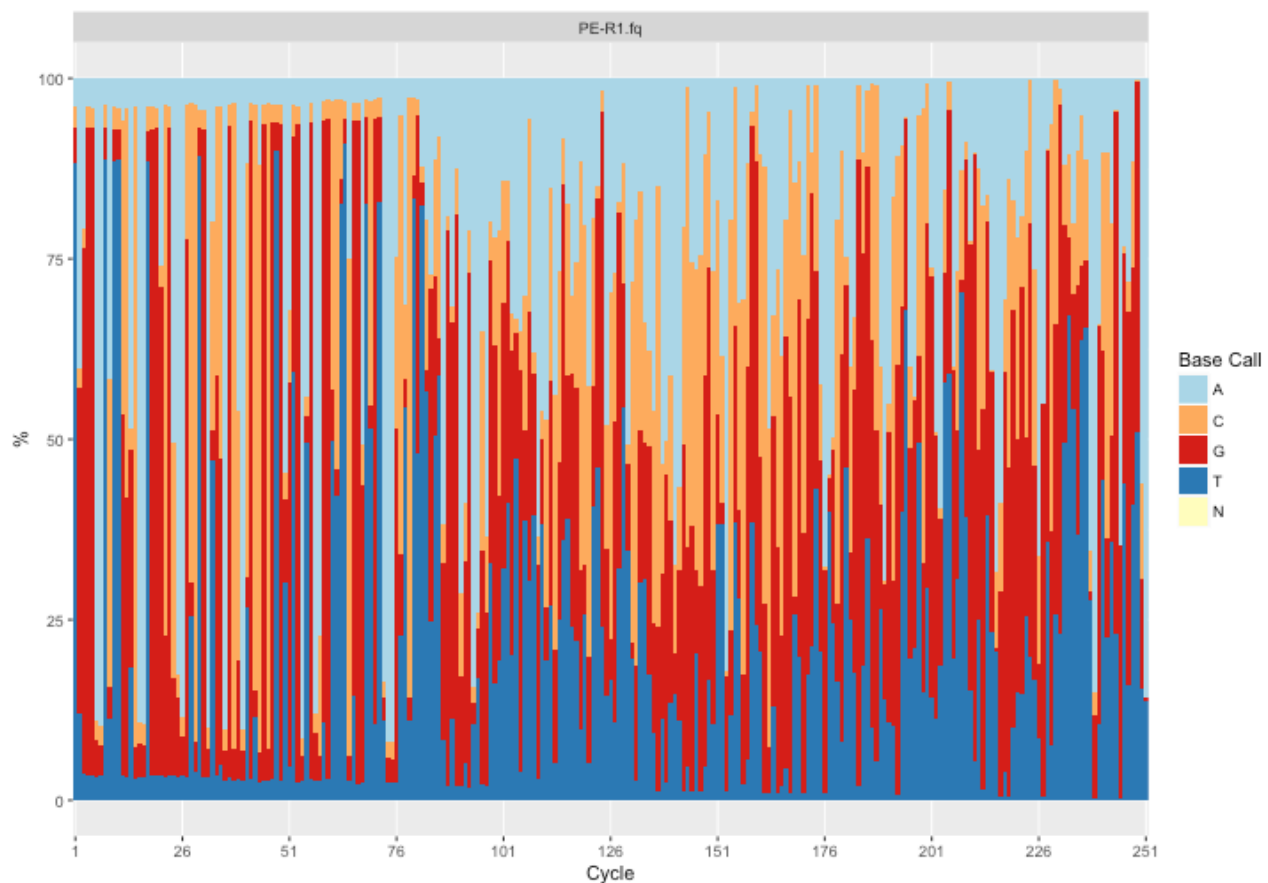


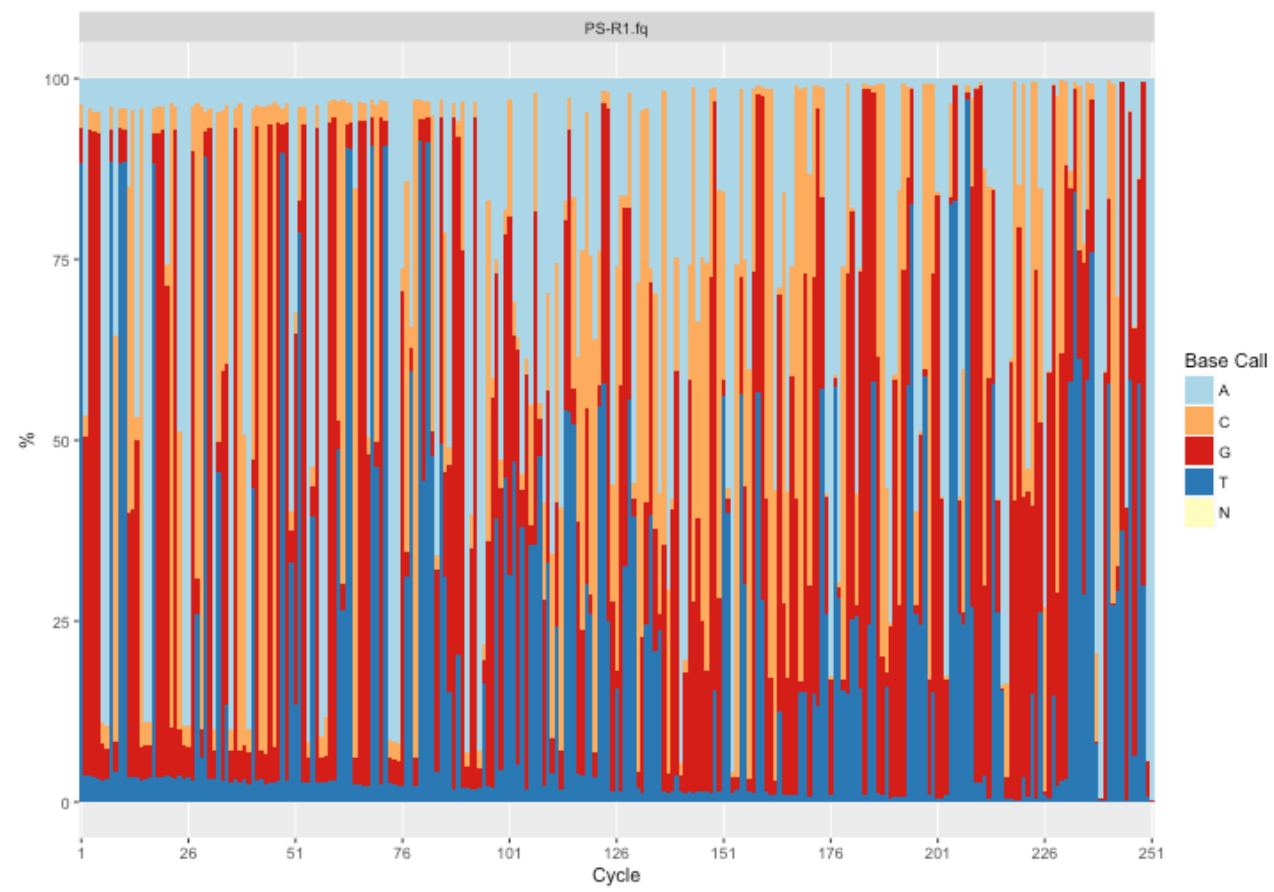
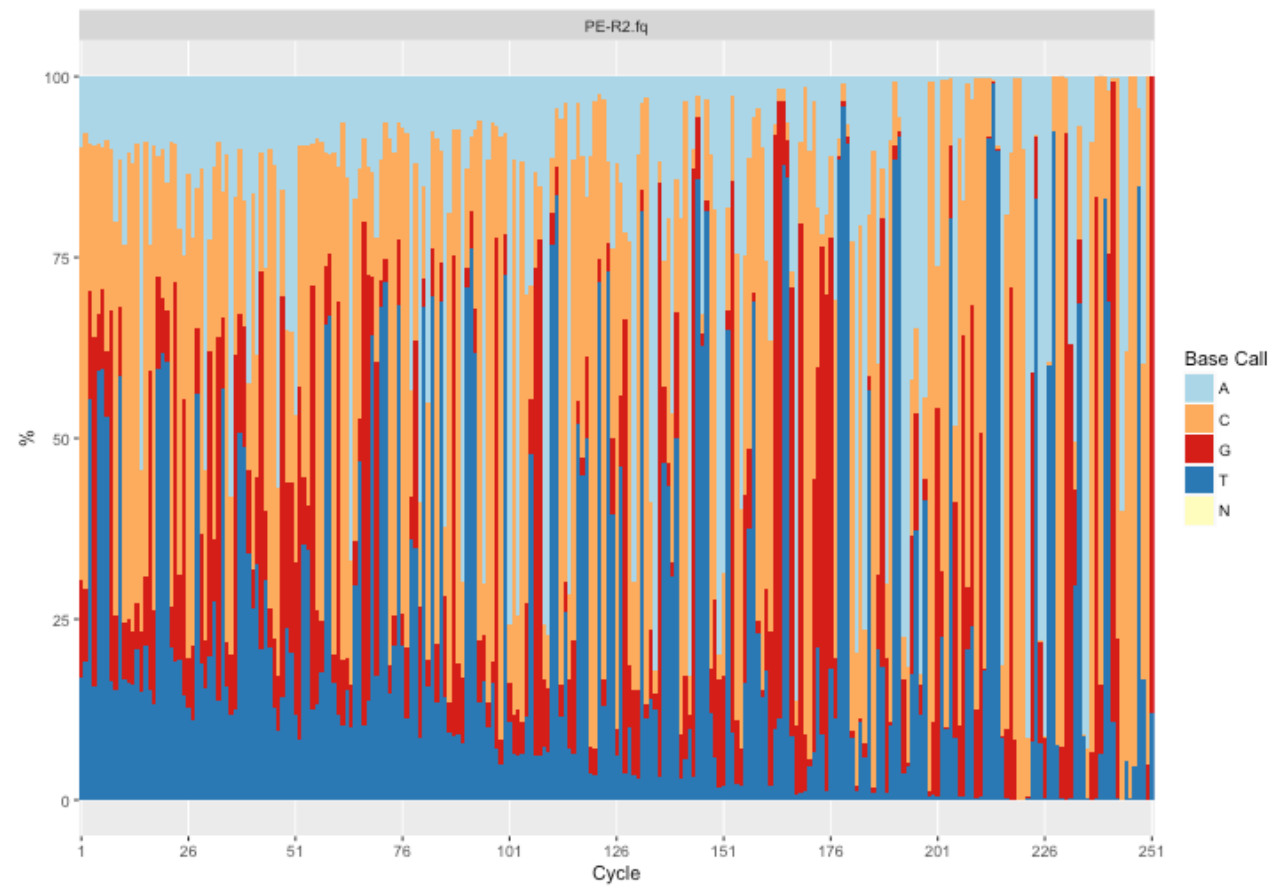


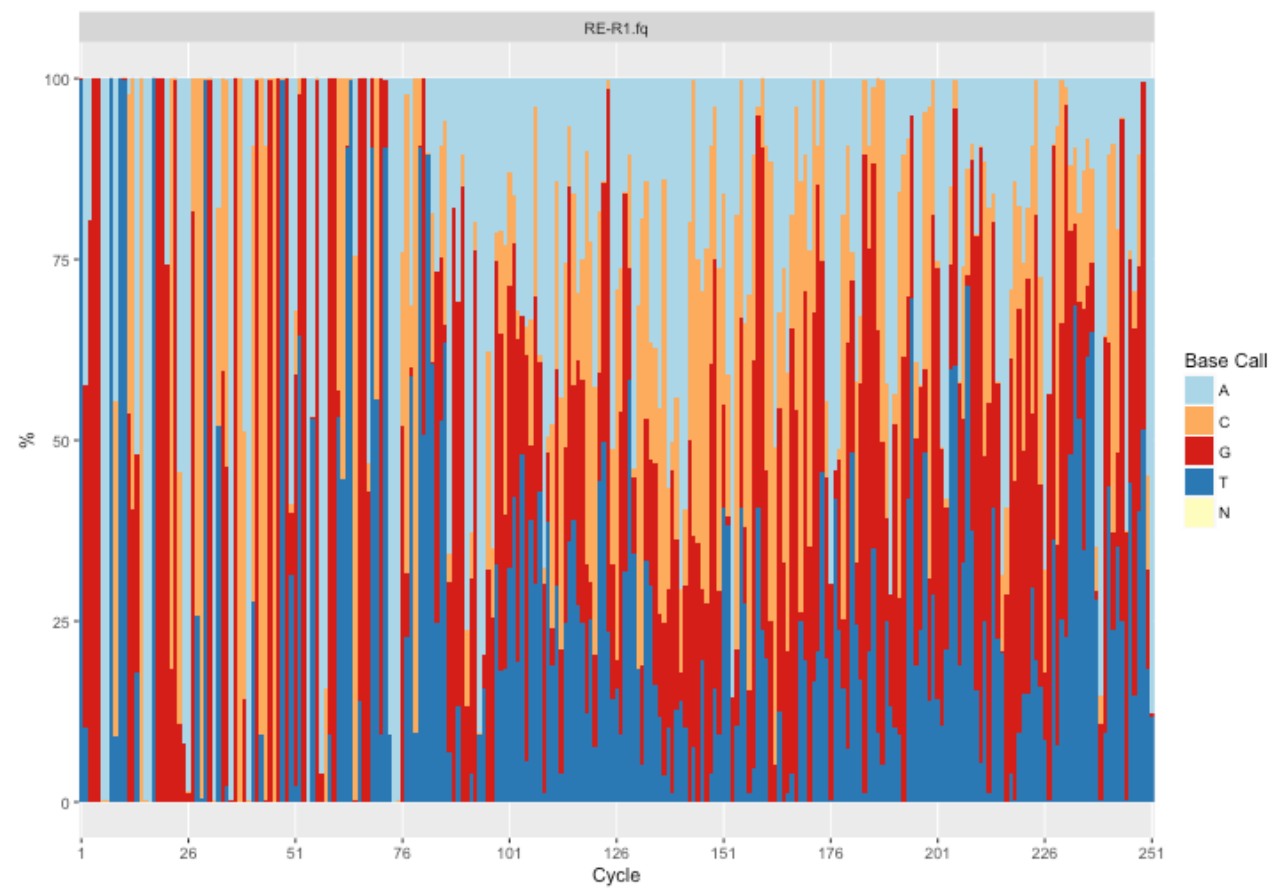
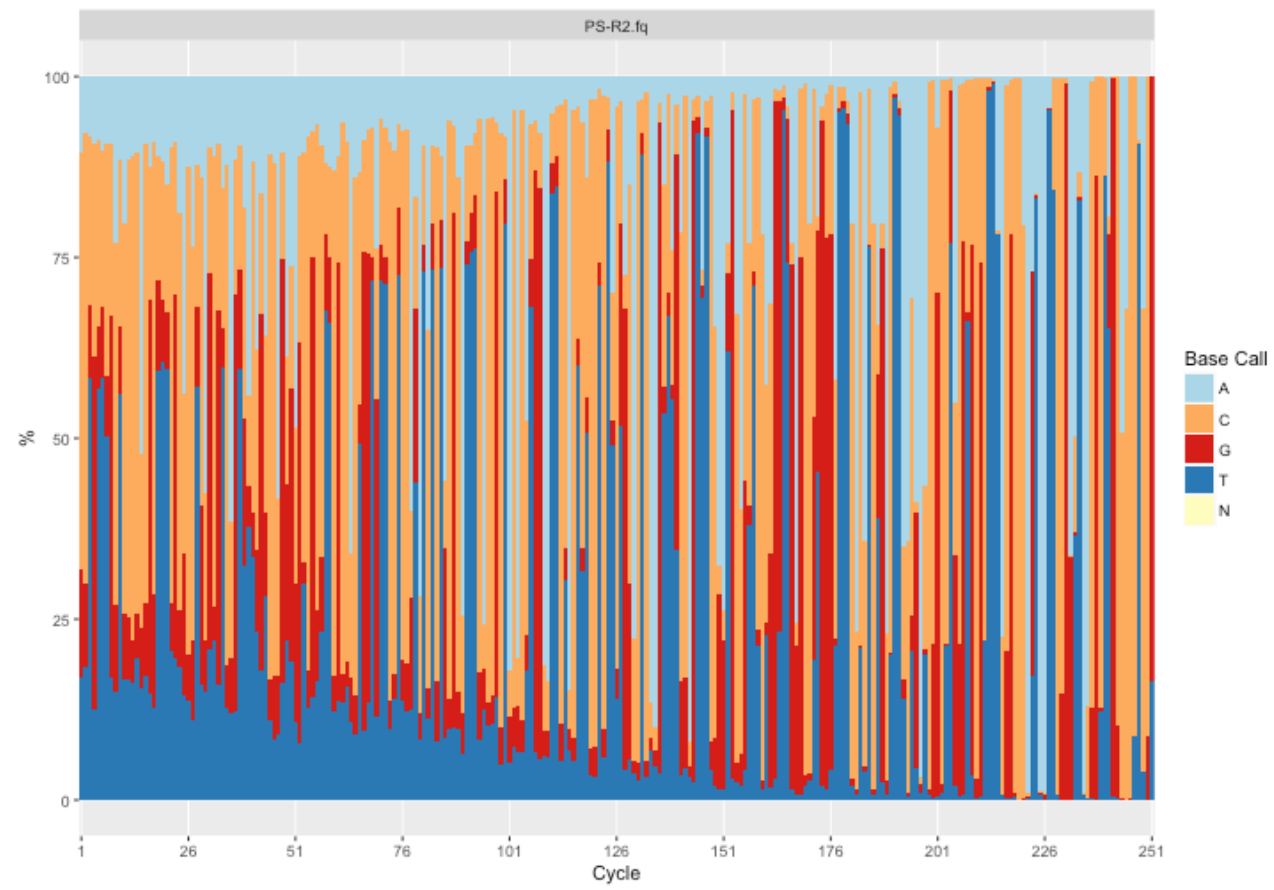
Cycle-specific Base Call Proportion

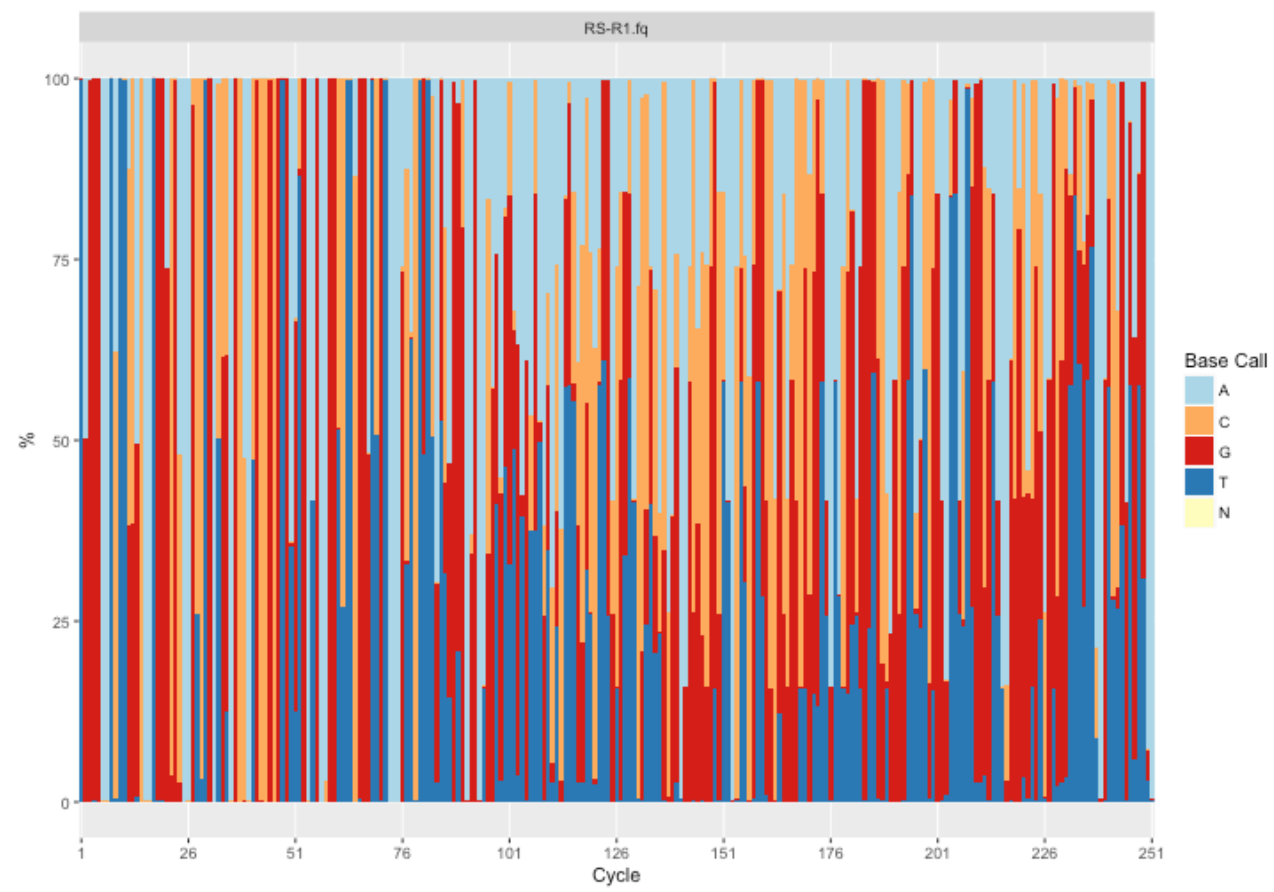
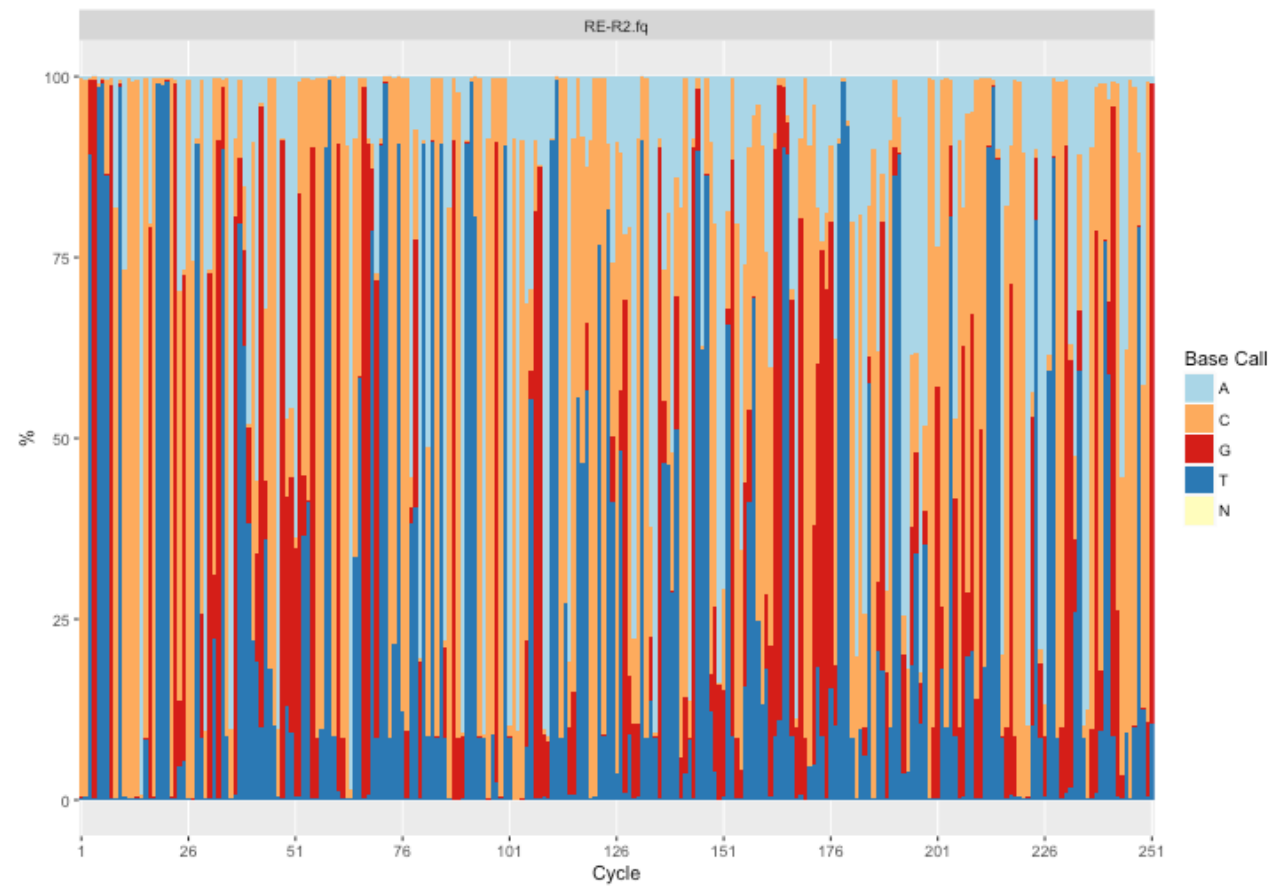
This stacked bar plot describes the proportion of each nucleotide called for every cycle of sequencing.

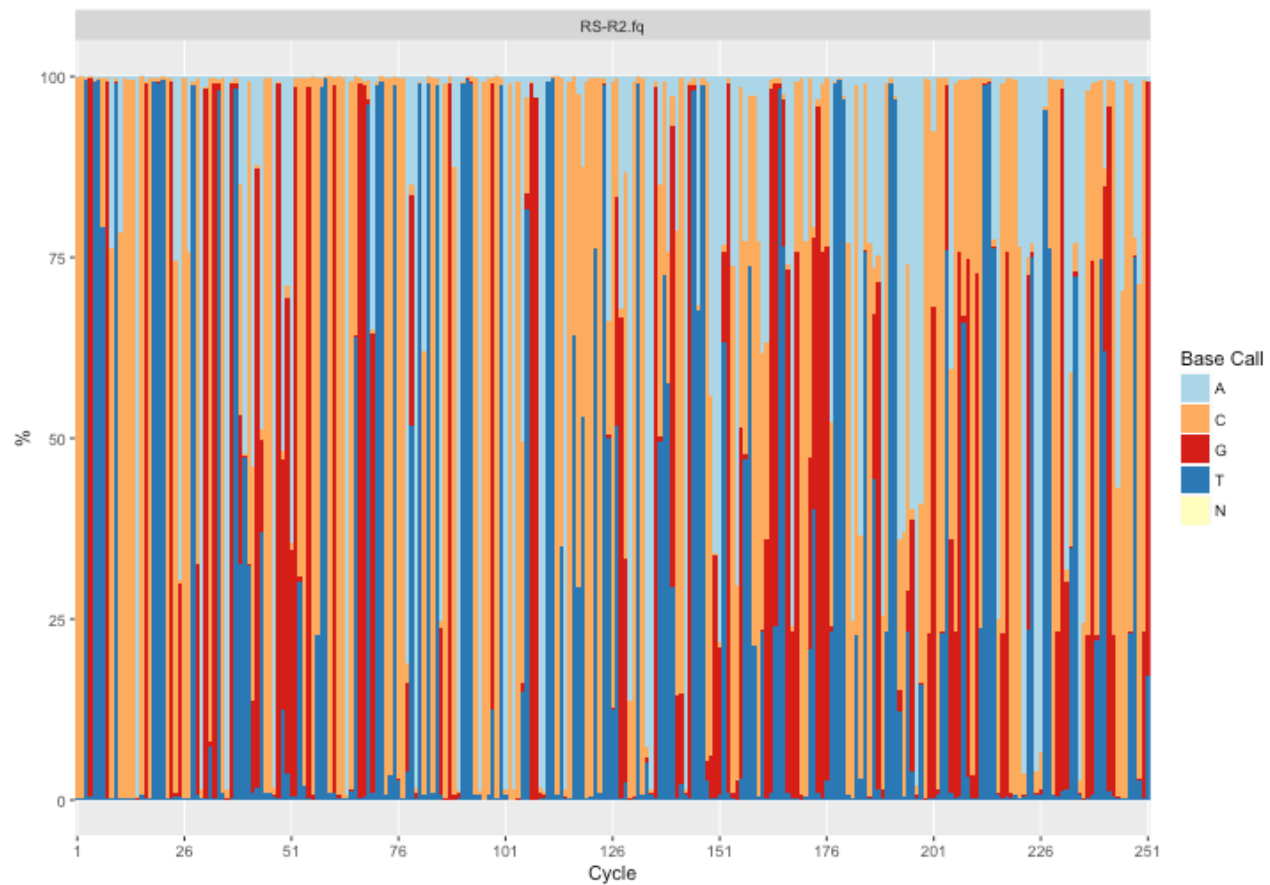
```
for(pair in pairs) {
  rqcResultSet.sub <- subsetByPair(rqcResultSet, pair)
  print(rqcCycleBaseCallsPlot(rqcResultSet.sub))
}
```











The line plot shown below contains the same information as the plot above. However, some may find this easier to read when comparing the calling rates for each of the nucleotides.

```
for(pair in pairs) {
  rqcResultSet.sub <- subsetByPair(rqcResultSet, pair)
  print(rqcCycleBaseCallsLinePlot(rqcResultSet.sub))
}
```

